European Commission

Research and Innovation

The Futures of Artificial Intelligence: Implications for Europe's R&I Ecosystem

Part I: A Taxonomy for Artificial Intelligence: Technology Trends, Capabilities and Applications



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Part I: A Taxonomy for Artificial Intelligence: Technology Trends, Capabilities and AI Applications

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EUROPEAN COMMISSION

The Futures of Artificial Intelligence: Implications for Europe's R&I Ecosystem

Part I: A Taxonomy for Artificial Intelligence: Technology Trends, Capabilities and AI Applications

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Foresight On Demand Consortium

Directorate-General for Research and Innovation

2025

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1. Introduction

Purpose and scope of the report

The purpose of the "Futures of AI: Implications Europe's R&I Ecosystem and EU Policy" study (the Study), which is requested under the 'Foresight on Demand' (FOD) framework contract, is to explore plausible futures for global Artificial Intelligence (AI) developments in Europe's industry and R&I ecosystem. The findings are expected to inform and generate recommendations for the EU's R&I policy in relation to AI and feed into the development of AI in Science strategy and support the potential setup of a European AI Research Council[1].

More specifically, the study will:

1. Explore the state of the art and future trends in AI development.
2. Estimate the current degree of use and development of AI across all European industries and the R&I ecosystem.
3. Explore AI's potential impact on the EU's R&I ecosystem and industrial sectors.
4. Provide an analysis of strengths, weaknesses, opportunities and threats for the EU R&I ecosystem.
5. Seek to identify key R&I policy actions for the EU and EU Member States.

The time horizon of the foresight **extends to the year 2040**.

The objective of this report, which constitutes the first step of the study, is to:

- Collect information on current trends, drivers and barriers in terms of the development and uptake of AI technology in Europe. When considering AI technological developments and uptake, the report provides a global overview as well.
- Develop a taxonomy on AI technologies. The developed taxonomy will be used to map the current state of AI in the European industrial landscape.

The key research questions that are covered in this report are described in the box below.

[1] Mission Letter for Commissioner-designate for Startups Research and Innovation (europa.eu)

Trends and challenges for AI technology generation:

- What are the recent key trends in AI technology developments from a supply-side perspective?
- What are the trends related to the cross-fertilisation of AI technologies with other technologies?
- What are the emerging AI technology areas?
- What are the technology maturity and societal readiness level of those AI technologies?
- What are the key drivers and barriers, needs and challenges, related to AI technologies generation in Europe?

Trends and challenges for AI technology adoption across industry sectors and application areas

- What are the key developments related to AI-technology development and uptake across the different industrial sectors and application areas? What is their potential transformative impact in those sectors?
- What are the key drivers and barriers, needs and challenges from industry for the adoption of AI technologies (demand side)?
- What is the market structure and trends related to the generation and adoption of AI-technologies at global scale (incl. world leaders in AI)?

Taxonomy building

- What are the existing taxonomies on AI technology development and uptake across sectors?
- What are the keywords and definition that can be used to cover the full scope of AI technologies and their applications across a broad range of relevant application sectors?

Text Box 1 - Key Research Questions

Methodological approach

The report is based on the review of literature and stakeholder consultation with the objective to 1) capture the trends in AI development and uptake; 2) identify key drivers and barriers; 3) build a taxonomy of AI. The reviewed sources covered the EU policy context as well as key global market, industry and technological trends across application areas. The list of consulted sources is available in the annex 3.1. The list of consulted stakeholders is available in the annex 3.3.

To develop a taxonomy of AI, multiple sources of information were analysed, including: 1) foresight literature reports; 2) portfolios of leading RTOs and research institutes; 3) Strategic Research and Innovation Agendas of European Partnerships; 4) ERC and EIC project portfolio; 5) Existing taxonomies. The taxonomy development process is visualised in the figure below.

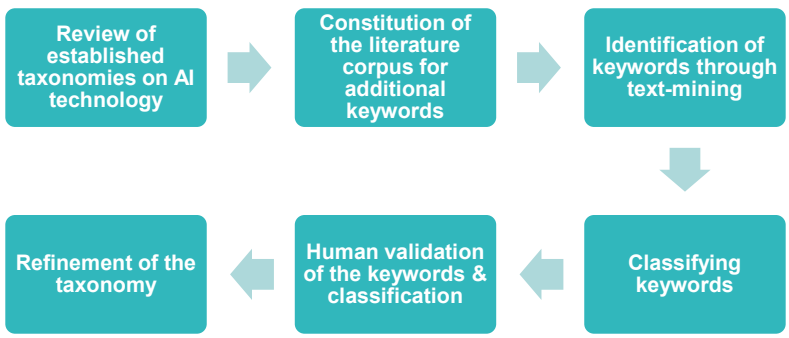


Figure 1 - Taxonomy Development Process. Source: Technopolis Group (2025)

A more detailed description **of the methodological approach towards taxonomy building** and the **use of the taxonomy further in the Study** is available in annex 3.2.

The AI technology trends, and preliminary taxonomy of AI have been **discussed and validated** with stakeholders **during a dedicated workshop session**, with technology and AI policy experts. The objectives of the workshop were to:

- Discuss and validate the key contextual factors influencing the development and uptake of Artificial Intelligence in the EU, including relevant policy and regulatory frameworks, as well as broader enabling conditions and barriers.
- Discuss and refine the taxonomy for Artificial Intelligence, encompassing both technological trends and application domains.
- Extend the AI taxonomy by examining technology maturity, societal readiness, and the "hype factor", together with the potential transformative effects of AI-based applications on industrial sectors and society.

2. Context and Trends in AI technology development & application sectors

There is no universal agreement on the definition of AI and various interpretations exist. One of the potential explanations of the lack of consensus is the broad range of tasks and capabilities AI systems can perform[2]. For example, Legg and Hutter[3] presented 70 definitions in 2007, Russell and Norvig presented eight definition approaches in 2012. The European Commission has also adopted a definition of AI: "'*AI system means a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments*"[4]. This definition comprises seven main elements: (1) a machine-based system; (2) that is designed to operate with varying levels of autonomy; (3) that may exhibit adaptiveness after deployment; (4) and that, for explicit or implicit objectives; (5) infers, from the input it receives, how to generate outputs (6) such as predictions, content, recommendations, or decisions (7) that can influence physical or virtual environments.[5]

2.1. Policy Context.

2.1.1. EU Level

Over the past years, the European Union has made significant steps to become a key strategic player in the field of AI (see figure below). Such efforts are reflected in several EU initiatives, including the Digital Decade Strategy, the European Data Strategy, as well as the AI Act, the Data Act, and the Digital Markets Act. The second von der Leyen Commission has also included

[2] Retrieved from: https://www.nasa.gov/what-is-artificial-intelligence/
[3] Shane Legg, Marcus Hutter (2007). Universal Intelligence: A Definition of Machine Intelligence. *Minds & Machines*, 17:4, pages 391-444.
[4] Retrieved from: https://artificialintelligenceact.eu/article/3/
[5] Retrieved from: https://digital-strategy.ec.europa.eu/en/library/commission-publishes-guidelines-ai-system-definition-facilitate-first-ai-acts-rules-application

"turning Europe into the AI continent" among its objectives. Additionally, the AI policy framework includes:

- The European AI Strategy – aims at making the EU a world-class hub for AI and ensuring that AI is human-centric and trustworthy.
- AI innovation package – includes several measures to support European startups and SMEs in the development of trustworthy AI that respects EU values and rules. One key element of this package is the Communication on boosting startups and innovation in trustworthy artificial intelligence that sets out a strategic investment framework in trustworthy AI for the Union to capitalise on its assets, in particular its world-leading supercomputing infrastructure, and to foster an innovative European AI ecosystem.
- AI Act – the objective of the Act is to address the risks generated by specific uses of AI through a set of complementary, proportionate, and flexible rules.

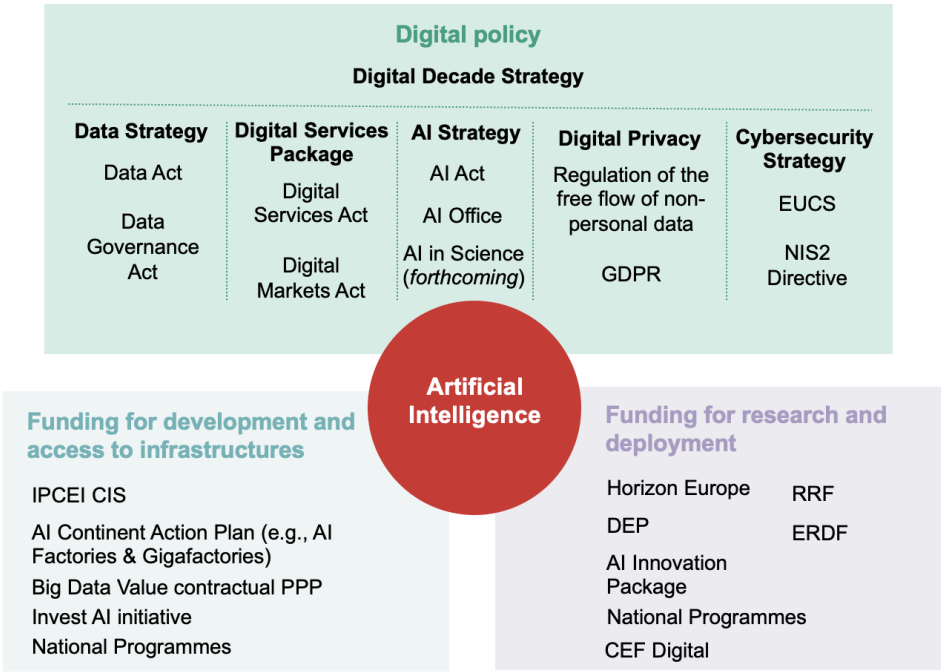


Figure 2 - Overview of the AI policy framework. Source: Technopolis Group (2025)

In terms of funding initiatives for research and deployment of AI, for example, under the Horizon Europe **Destination 3 (World Leading Data and Computing Technologies**), key topics encompass AI-driven data operations, compliance technologies, and efficient, trustworthy AI for data optimisation. Projects like AI-DAPT focus on automating AI pipelines with human oversight and hybrid science-guided models[6]. **Destination 4 (Digital & Emerging Technologies for Competitiveness and Fit for the Green Deal)** emphasises AI-driven autonomous robots, advanced human-robot interaction, and novel AI paradigms for robotics. **Destination 6 (Human-Centric and Ethical Development of Digital and Industrial Technologies)** prioritises ethical AI, reliable data optimisation, and AI pipeline automation, ensuring trustworthiness and

[6] Retrieved from: https://www.ai-dapt.eu/project/

alignment with European values[7]. AI is also addressed across **Pillar I and other Pillar II Clusters and Pillar III**. For example, the figure below provides an overview of ERC (Pillar I) AI project portfolio from FP7 to Horizon Europe. Overall, ERC-funded AI projects represent around 7% of all ERC-funded projects from 2007 up to 2022.

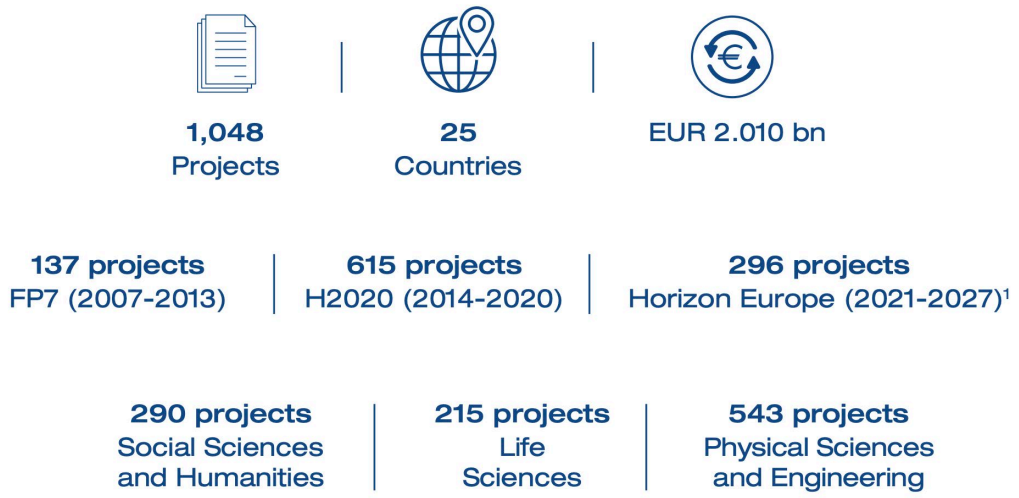


Figure 3 ERC AI project portfolio. Source: ERC (2024), Mapping ERC Frontier Research Artificial Intelligence. Retrieved from: https://erc.europa.eu/sites/default/files/2024-03/report-AI.pdf

In addition, the **European High Performance Computing Joint Undertaking (EuroHPC)** has selected seven proposals to establish and operate **AI Factories across Europe**. AI Factories are dynamic ecosystems that foster innovation, collaboration, and development in the field of AI. They bring together computer power, data, and talent to create cutting-edge generative AI models[8].

In terms of regional funding, a recent study by JRC found that during 2014-2020, around EUR 8 billion of EU funds were targeted for AI investments in the European regions. More developed regions have a higher specialisation in AI EU-funded investments. AI-related investments are more concentrated in regions with a higher concentration of ICT activities and that are more innovative, highlighting the importance of agglomeration effects.[9]

[7] Retrieved from: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/wp-call/2023-2024/wp-7-digital-industry-and-space_horizon-2023-2024_en.pdf
[8] Retrieved from: https://digital-strategy.ec.europa.eu/en/policies/ai-factories
[9] Santos, Anabela Marques, Francesco Molica, and Carlos Torrecilla Salinas. "EU-funded investment in Artificial Intelligence and regional specialization." Regional Science Policy & Practice (2025): 100190.

2.1.2. EU Member States level

Figure 4 provides an overview of AI policy initiatives introduced from 2010 to 2023 in the EU-27.

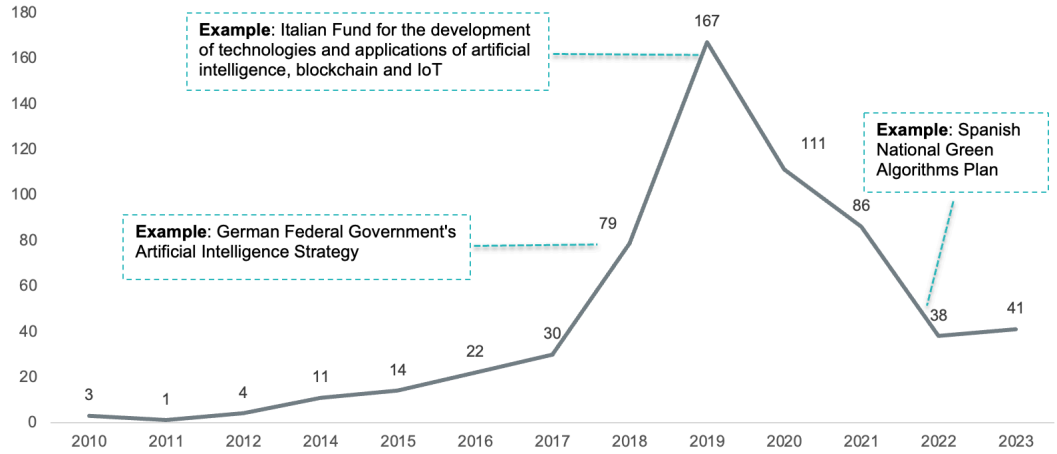


Figure 4 - AI policy initiatives introduced each year from 2010 to 2023. Source: Technopolis Group based on OECD AI policy observatory.

The **AI policy initiatives saw exponential growth from 2015 to 2019**. After 2019, the pace of new initiatives slowed. According to OECD data, a sharp rise in AI policy initiatives started in 2017 (30 initiatives) and peaked in 2019 (167 initiatives). **This period likely reflects increasing EU interest and concern in AI regulation, ethics, and governance as AI technologies advanced rapidly**. Some examples of the initiatives include:

- German Federal Government's AI Strategy (2018): The goal is to establish "AI made in Germany" as an international trademark for cutting-edge, secure AI applications aimed at serving the common good in line with Europe's core values[10].
- Italian Fund for the development of technologies and applications of AI, blockchain and IoT (2019): The objective is to finance experimentation and testing activities of the above-mentioned emerging technologies. The Fund has had a budget of €45 million for the 2019-2021 period.[11]
- Spanish National Green Algorithms Plan (2022): The objective is to develop AI in an eco-sustainable and resilient way, in line with the objective of the EU of ensuring that digital infrastructure and technologies, including their supply chains, become more sustainable and resilient[12].

The portfolio of policy measures across the EU MS 27 is reflected in the figure below.

[10] Retrieved from: https://www.plattform-lernende-systeme.de/ai-strategies.html
[11] Retrieved from: https://stip.oecd.org/stip/interactive-dashboards/policy-initiatives/2023%2Fdata%2FpolicyInitiatives%2F23975
[12] Retrieved from: https://depp.oecd.org/policies/ESP2633

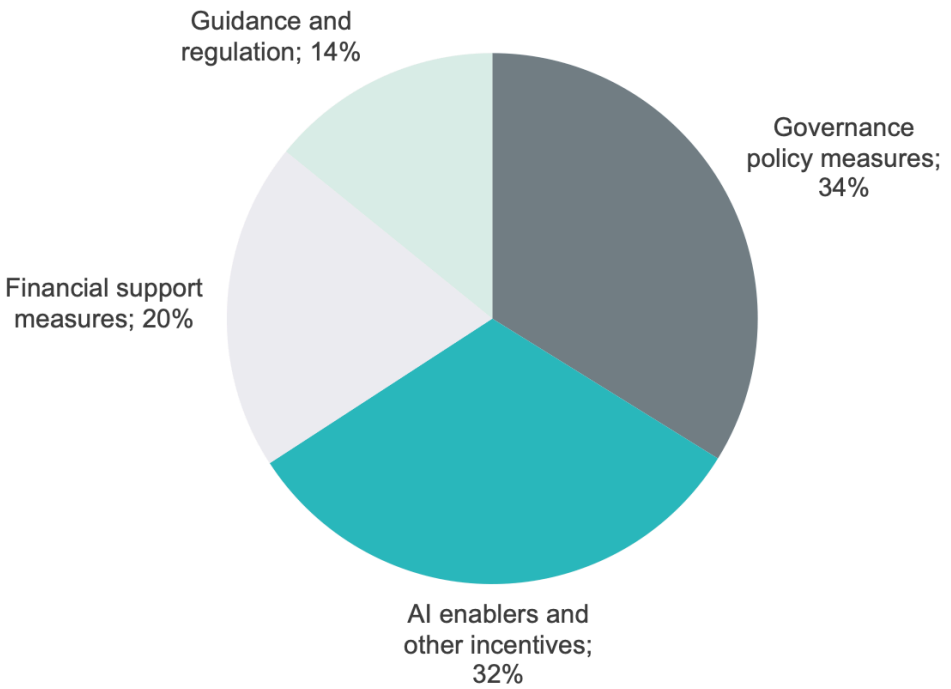


Figure 5 - Portfolio of policy measures related to AI across the EU MS 27 (all policy instruments from 2010 to 2023). Source: Technopolis Group based on OECD AI policy observatory.

In terms of key takeaways, **governance measures** (e.g., national/regional strategies) **and AI enablers**[13] are the dominant policy initiatives, representing 34% and 32% of all policy instruments respectively. **Financial support** (e.g., grants, institutional funding for public research, indirect financing, equity financing) is substantial and represents 20% of all policy instruments. **Guidance and regulation** (e.g. regulatory oversight and ethical advice bodies, standards and certification for technology development and adoption) are comparatively less frequent, representing 14% of all instruments.

[13] It includes: AI skills and education, labour market policies, public awareness campaigns, data access and sharing, AI computing and research, networking and collaboration.

2.2. AI technology generation trends

AI technology generation trends can be reflected by several indicators, including publications by field of study, patent applications and venture capital investments.

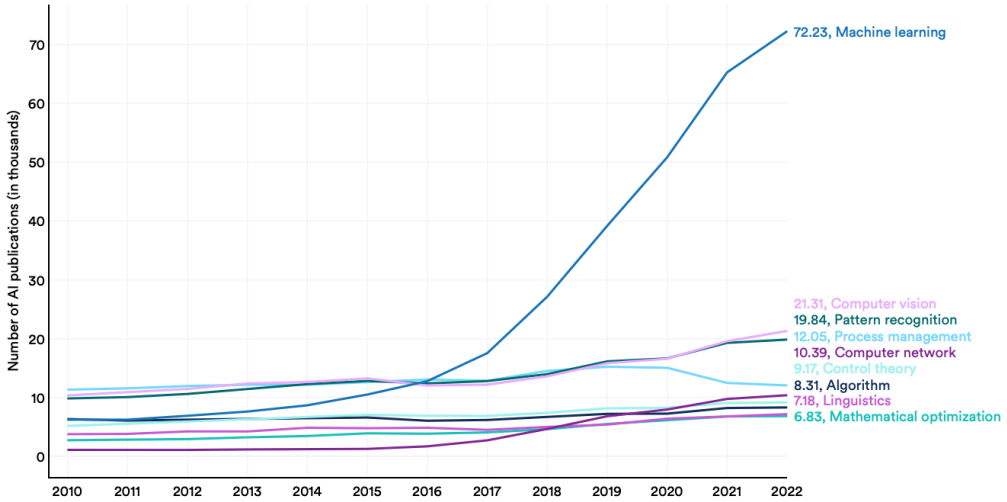


Figure 6 -Number of AI publications by field of study from 2010 to 2022. Source: Centre for Security and Emerging Technology, 2023

Machine learning publications have seen the most rapid growth over the past decade, increasing nearly sevenfold since 2015. Following machine learning, in 2022 researchers published the most in computer vision (21,309 publications), pattern recognition (19,841), and process management (12,052).

This trend in publications was also reflected by the Association for the Advancement of Artificial Intelligence report which called for encouraging and support research on a variety of AI paradigms. This includes diverse methodologies (beyond machine learning)[14].

To further identify the research trends in AI a literature review of academic publications, policy and market reports & studies was carried out (see annex 3.1 for consulted sources). The technological research and development trends in AI captured through desk research are summarised in the figure below. In short, these trends encompass the core technical components of AI – (Machine learning (including neural networks), NLP and key AI components), aspects related to the use of AI (Governance & Ethics & Safety) and how and where the AI can be used (AI capabilities and application areas).

[14] Retrieved from: https://aaai.org/wp-content/uploads/2025/03/AAAI-2025-PresPanel-Report-FINAL.pdf

Machine Learning, NLP and key AI components	Governance & Ethics & Safety	AI capabilities & application areas
• Feature Engineering • Machine learning • Natural Language Processing • Networks • AI Systems and Platforms	• AI Safety & Risk Management • AI Ethics & Fairness • AI Governance & Policy	• Emotion & Face regognition and creation • Image & object regognition and creation • Speech & voice regognition and creation • Forecasting, planning, recommending • Autonomous systems (ability to change behaviour in response to unexpected events / change • AI integration / interaction with other systems and platforms across private and public sector and use cases

Table 1 - High-level overview of the key AI technological trends. Source: Technopolis Group (2025)

Overall, the field of AI has simultaneously encompassed many different methodologies and research paradigms. There is a risk that the current convergence of the field towards focusing on neural approaches could impede innovation. Consequently, an active support of research on classical (non-neural) approaches, as well as research that combines neural approaches with other approaches, and that integrates various paradigms into more complete cognitive architectures could yield interesting results. In addition, the support of creative investigations of completely new paradigms may be the key to overcoming the limitations of existing paradigms.[15]

2.2.1. AI models.

Another indicator that captures the developments in AI is the number of new AI models introduced. It should be noted that many new AI models are introduced each year, but only a few are the most advanced models. According to EpochAI, an advanced AI model has to meet one of the following criteria: (i) state-of-the-art improvement on a recognised benchmark; (ii) highly cited (over 1000 citations); (iii) historical relevance; (iv) significant use[16]. The figure below displays the global trend of the most advanced machine learning models developed by different types of organisations.

[15] Retrieved from: https://aaai.org/wp-content/uploads/2025/03/AAAI-2025-PresPanel-Report-FINAL.pdf
[16] Retrieved from: https://epoch.ai/data/notable-ai-models#faq

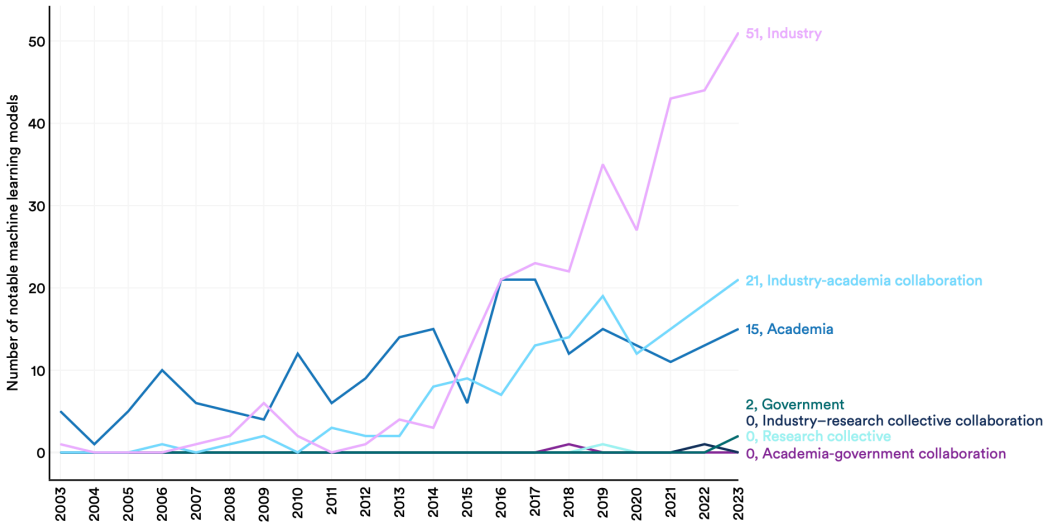


Figure 7 - Number of notable machine learning models by different types of organisations. Source: Epoch, 2023 & AI Index report

The figure shows that until 2014, academia led in terms of introducing new advanced machine learning models. Since then, the industry has taken the lead. In 2023, there were 51 notable machine learning models produced by industry compared to just 15 from academia. **This shift toward industry-led AI technology development is likely driven by the fact that creating cutting-edge AI models now requires vast amounts of data, computing power, financial resources, and, crucially, talent – areas where academia struggles to compete with industry**[17].

Introduced in 2023, the Ecosystem Graphs is a community resource from Stanford that tracks global developments in foundation models, including datasets, specific models, and applications. The figure below, displays the number of foundation models introduced since 2019 by their access type.

[17] Retrieved from: https://epoch.ai/trends#compute

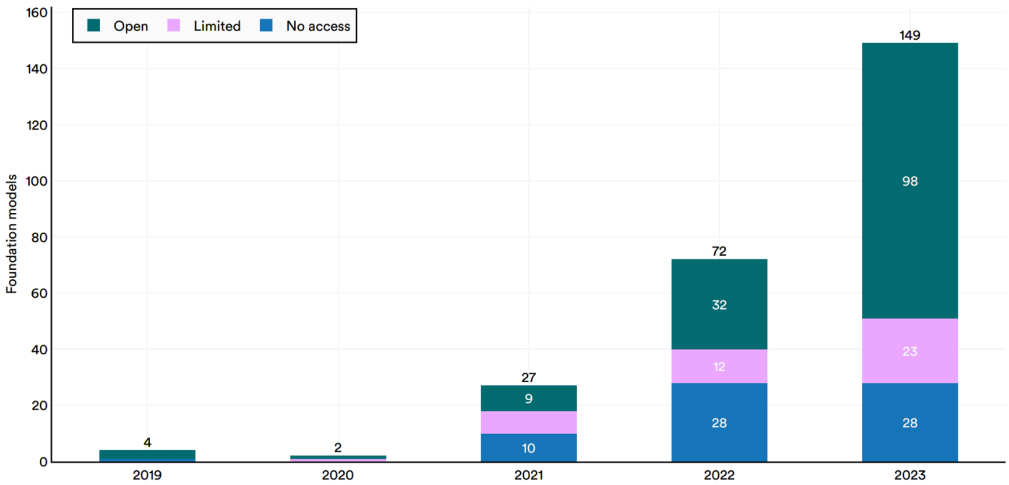


Figure 8 - Number of foundation models by access type since 2019. Source: AI Index report.

In recent years, the number of foundation models has risen sharply, more than doubling since 2022 and growing by a factor of nearly 38 since 2019. Out of 149 foundation models released in 2023, 98 were open, 23 provided limited access and 28 models were fully proprietary. Open-source and self-owned AI models significantly differ regarding aspiration, impact on creativity, and availability. For example, open-source Large Language Models (LLMs) are designed explicitly to share AI technology globally, making it available for researchers, developers, and organisations globally without restrictions[18]. Proprietary models are developed by organizations with the intent of improving their positioning in the market.[19] These models are usually proprietary or distributed through limited access Application Programming Interface (APIs), thus limiting the ability for outside contribution and inspection.

Both proprietary and open-source models contribute to AI technological advancement. However, while increasing dispersion of AI technology across different industries and fields is the advantages of open-source models, proprietary models are more suited for commercial outcomes and demonstrate state-of-the art technological advancements.

2.2.2. Patents

Patents filed in AI are one of the classical indicators capturing the technological development trends. According to a recent study prepared by World Intellectual Property Organization, **globally, machine learning is the dominant artificial intelligence technique**. Reference to machine learning is found in 40% of all AI-related patents analysed and the technique grew at an average rate of 28% every year from 2013-2016[20]. Patenting activity in Generative Artificial Intelligence (GenAI) indicates that it has also received significant attention from public and private organisations. The figure below provides an overview of the AI patent trends by core GenAI models.

[18] Retrieved from: https://wjaets.com/sites/default/files/WJAETS-2024-0562_0.pdf
[19] Retrieved from: https://wjaets.com/sites/default/files/WJAETS-2024-0562_0.pdf
[20] Retrieved from: https://www.wipo.int/en/web/technology-trends/artificial_intelligence/story

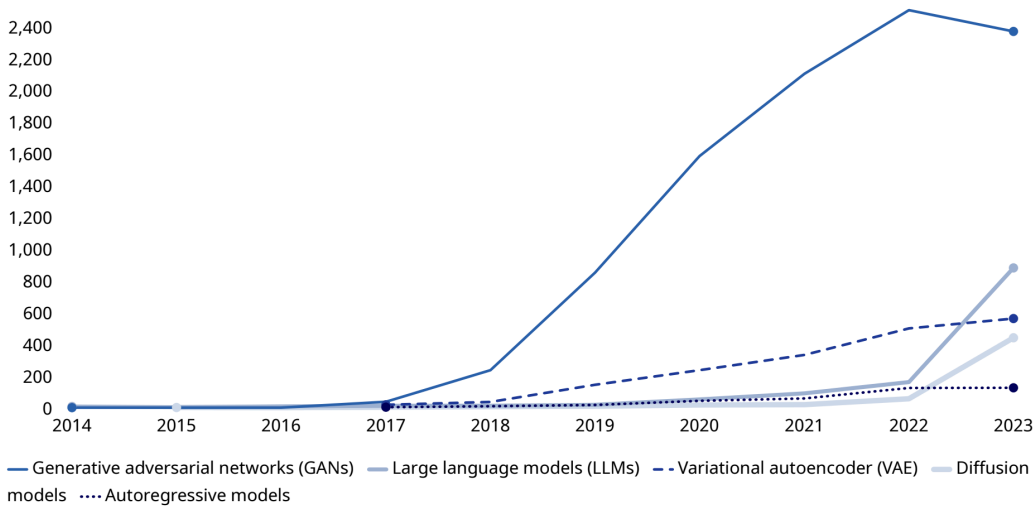


Figure 9 - AI patent trends by core GenAI models. Source: AI Index report

Among the core models, many **patent families belong to the category of generative adversarial networks** (GANs). Between 2014 and 2023, around 9700 patent families in this model have been published. Variational autoencoders (VAE) and LLMs are the second and third largest models in terms of patent families with 1800 and 1300 new patent families respectively between 2014 and 2023.

In terms of AI applications, computer vision, which includes image recognition (critical for self-driving cars), is the most popular application of AI referenced in patent documents. It was mentioned in 49% of all AI-related patents and grew annually at an average rate of 24% over the period 2013-16[21].

2.2.3. Private Investments

Venture capital and private equity investments serve as another key indicator of technology generation in the field of AI. Venture capital and private equity tend to finance companies that are not quoted on the stock market and are often in the early development phase. These investments can act as a proxy to help understand which countries, industries or technologies are attracting investments. **Until 2021, global private investment in AI have been gradually increasing since 2013**[22]. Nevertheless, since 2021 global private AI investment has declined for the second consecutive year (see figure below).

[21] Retrieved from: https://www.wipo.int/en/web/technology-trends/artificial_intelligence/story
[22] Retrieved from: https://www.mediahuis.ie/app/uploads/2024/04/HAI_AI-Index-Report-2024.pdf

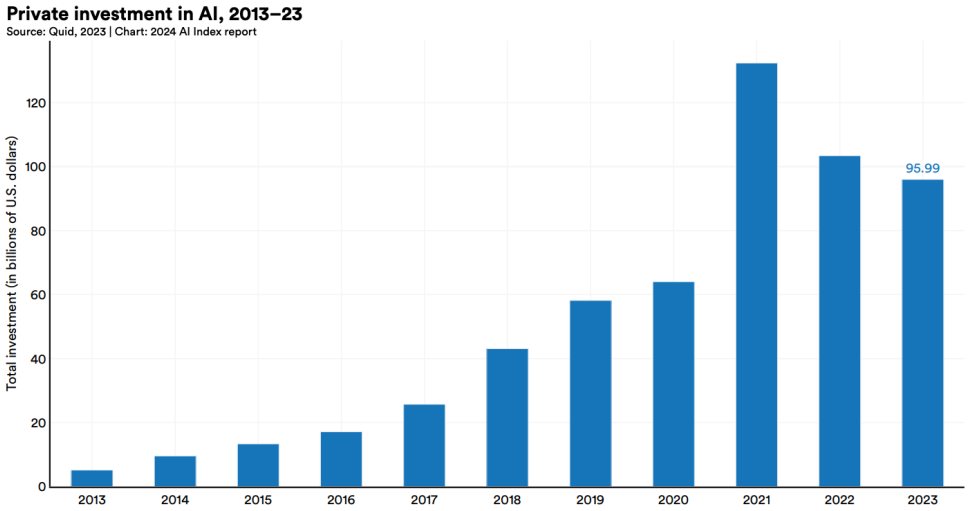


Figure 10 - Global private investments in AI. Source: AI Index report

2.2.4. Hardware trends

In terms of specific hardware technology trends, the availability of data plays a crucial role in driving AI model development. The most powerful AI systems to date are language models that are primarily trained on trillions of words of human-generated text from the internet. However, there is only a finite amount of publicly available human-generated data. According to EpochAI research[23], the total stock of human-generated public text data is 300 trillion tokens. This estimate includes only data that is sufficiently high-quality to be used for training and accounts for the possibility of training models for multiple epochs. EpochAI **forecasted that this data stock will be fully utilised at some point between 2026 and 2032** considering the historical growth rate in dataset sizes and projection of training compute growth[24].

Another trend that has implications for the development of AI models is the shortage of AI chips. AI chips provide the compute capacity necessary for training large AI models. Bain & Company research showed that **businesses should expect a likely AI chip shortage, with AI driving demand for chips components by 30% or more by 2026**[25]. A demand increase of 20% or more would put a damper on the AI chip supply chain[26].

Finally, addressing the **data centre and cloud capacity gap** is crucial, as Europe's data centre capacity needs to triple by 2027, reaching approximately 22,700 MW to support AI development[27]. This shortfall can have significant implications for Europe's ability to achieve technological sovereignty, maintain competitiveness, and accelerate AI advancements.

[23] Retrieved from: https://epoch.ai/research
[24] Retrieved from: https://epoch.ai/research
[25] Retrieved from: https://www.bain.com/insights/prepare-for-the-coming-ai-chip-shortage-tech-report-2024/
[26] Retrieved from: https://www.bain.com/insights/prepare-for-the-coming-ai-chip-shortage-tech-report-2024/
[27] Retrieved from: https://pdf.euro.savills.co.uk/european/european-commercial-markets/spotlight-european-data-centres---may-2024.pdf

2.3. AI technology adoption trends and sectoral applications

Globally, AI adoption has spiked over the past five years. The latest McKinsey report reveals that in 2023, 55% of organisations surveyed have implemented AI in at least one business unit or function, marking a slight increase from 50% in 2022 and a significant jump from 20% in 2017.

In 2024, 13.48% of enterprises in the EU (an increase of 5.45 percentage points compared to 2023), with 10 or more employees and self-employed persons, used at least one of the following AI technologies:[28]

- Technologies analysing written language (text mining).
- Technologies converting spoken language into a machine-readable format (speech recognition).
- Technologies generating written or spoken language (natural language generation).
- Technologies identifying objects or people based on images (image recognition, image processing).
- Machine learning (e.g. deep learning) for data analysis.
- Technologies automating different workflows or assisting in decision-making (AI based software robotic process automation).
- Technologies enabling machines to physically move by observing their surroundings and taking autonomous decisions.

In some economic activities AI is used a lot more than in others. **This indicates that AI is more relevant for certain activities**. Globally, when considering the types of AI capabilities integrated into at least one function or business unit, robotic process automation had the highest adoption rate within the financial services industry (46%), as shown in the figure below. Virtual agents followed closely, also within financial services. Across all industries, the most widely embedded AI technologies were natural language text understanding (30%), robotic process automation (30%), and virtual agents (30%).

[28]Retrieved from: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Use_of_artificial_intelligence_in_enterprises

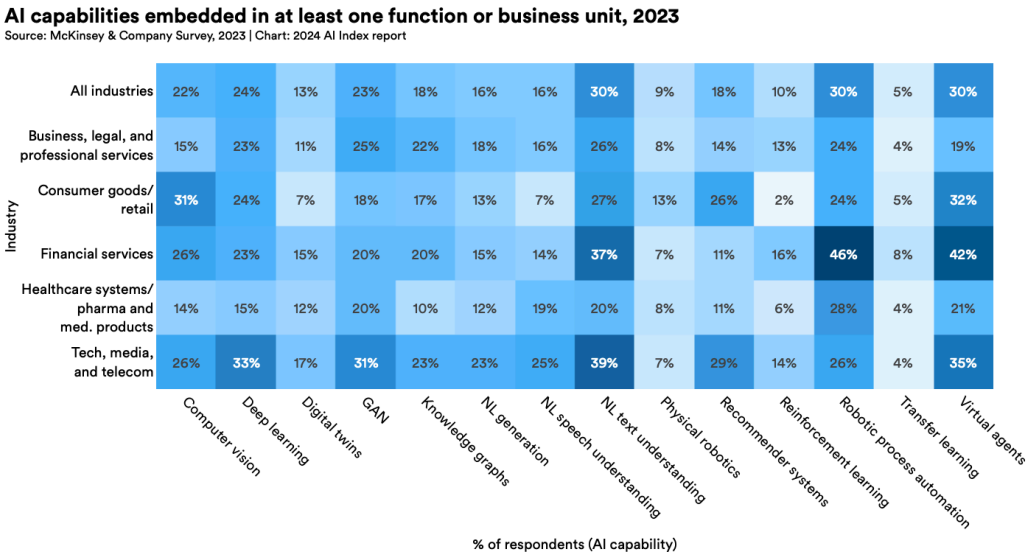


Figure 11 - AI capabilities embedded in at least one function or business unit in 2023. Source: AI Index report.

The table below presents the different types of AI technologies used across different economic activities in the EU. **In the information and communication sector, where the highest share of enterprises using AI was recorded, the most used AI technologies were text mining** (30.11%), followed by **natural language generation** (25.83%) and **machine learning for data analysis** (25.66%). In professional, scientific and technical service activities, text mining was again the most used AI technology with 15.61% of enterprises, followed by speech recognition (12.49%) and natural language generation (11.51%). In all other activities, the shares of enterprises using specific AI technologies ranged from less than 1% to 8%.

Activity	Performing analysis of written language (text mining)	Generating written or spoken language (natural language generation)	Converting spoken language into machine-readable format (speech recognition)	Machine learning (e.g. deep learning) for data analysis	Automating different workflows or assisting in decision making	Identifying objects or persons based on images (image recognition, image processing)	Enabling physical movement of machines via autonomous decisions
Manufacturing	5%	4%	3%	3%	3%	3%	1%
Electricity, gas, steam and air conditioning supply; water supply; sewerage, waste management and remediation activities	6%	4%	4%	5%	5%	3%	1%
Construction	3%	2%	3%	1%	1%	1%	0%
Wholesale and retail trade; repair of motor vehicles and motorcycles	6%	5%	4%	3%	3%	3%	1%
Retail trade, except of motor vehicles and motorcycles	5%	5%	3%	3%	3%	3%	1%
Transportation and storage	4%	3%	3%	2%	2%	2%	1%
Accommodation	3%	2%	2%	1%	1%	1%	0%
Information and communication	**30%**	**26%**	**20%**	**26%**	**21%**	**14%**	**3%**
Real estate activities	7%	6%	7%	3%	5%	2%	1%
Professional, scientific and technical activities	**16%**	**12%**	**12%**	**11%**	**10%**	**7%**	**2%**
Administrative and support service activities	8%	5%	5%	4%	4%	3%	1%
All activities	**7%**	**5%**	**5%**	**4%**	**4%**	**3%**	**1%**

Table 2 - Enterprises using AI technologies by type of AI technology and economic activity, EU, 2024. Source: Eurostat

As noted above, AI adoption varies significantly across economic activities, with some industries leveraging AI more extensively due to its greater relevance to their operations. **At the same time, certain sectors are better positioned for AI uptake**. To assess this, the OECD has developed an AI Intensity Sectoral Index, which combines indicators such as AI human capital (AI-related vacancies), AI innovation (AI-related patents), barrier-adjusted AI exposure (e.g., employees' abilities related to AI used within an occupation in industry), and AI usage. The figure below provides an overview of AI intensity across various sectors.

Sectors with high AI Intensity	Sectors with medium AI Intensity	Sectors with low AI Intensity
Computer & electronics	Chemicals	Food products
Media	Pharmaceuticals	Textiles & apparel
Telecommunications	Electrical equipment	Wood & paper
IT services	Machinery & equipment	Rubber, plastics, minerals
Finance & insurance	Transport equipment	Metal products
Legal & accounting	Other manufactures	Construction
Scientific R&D	Wholesale & retail	Hotels & food services
	Transportation & storage	
	Real estate	
	Other business services	
	Administrative & support services	

Table 3 - AI sectoral intensity. Source: OECD A sectoral taxonomy of AI intensity. Note: a sector is high if it ranks in the top quartile for at least two indicators and not in the bottom quartile for the remaining two and low if it ranks in the bottom quartile for at least two indicators and not in the top quartile for the remaining two.

Overall, **AI adoption remains consistently higher in the ICT and professional services sectors**, indicating that AI use is not yet evenly distributed across industries. These sectors benefit from complementary assets that facilitate AI adoption, including factors such as ICT skills, training, firm-level digital capabilities, and robust digital infrastructure. Given that AI is still in the early stages of diffusion, its full potential as a general-purpose technology broadly used across industries has yet to be fully realised. [29]

[29] Retrieved from: https://www.oecd.org/en/publications/a-sectoral-taxonomy-of-ai-intensity_1f6377b5-en.html

2.4. Drivers and barriers for the development and uptake of AI

The private and public sector faces several drivers and barriers related to the development and deployment of AI technologies. The level of importance was assessed consensus in expert judgement during the validation workshop. These elements are described in the tables below.

2.4.1 Drivers for the development and uptake of AI

Feature	Description	Level of importance
Drivers related to AI technologies generation and adoption in Europe		
Skills availability	The most advanced AI today cannot operate without humans. Availability and diversity in AI talent pools enhance creativity and problem-solving, driving AI development and adoption across industries. In general, organisations need AI developers and AI translators. These include people who know how to use and deploy gen AI, predictive analytics, LLMs, natural language processing (NLP), machine learning (ML), deep learning and reinforcement learning.[30] Moreover, not all AI-related skills demand extensive expertise in deployment. Some tasks require only foundational knowledge, such as prompt-tuning or fine-tuning ChatGPT. These skills are also linked to the broader digital competencies of the population. In the EU, for example, 56% of individuals aged 16 to 74 had at least basic digital skills in 2023.[31]	High
AI factories	AI Factories are dynamic ecosystems that foster innovation, collaboration, and development in the field of artificial intelligence. They bring together computer power, data, and talent to create state-of-the-art generative AI models. They foster collaboration across Europe, linking universities, supercomputing centres, industry, and financial actors. AI factories serve as hubs driving advancements in AI applications across various sectors.[32]	High

[30] Retrieved from: https://www.ibm.com/think/insights/ai-skills-gap
[31] Retrieved from: https://ec.europa.eu/eurostat/web/products-eurostat-news/-/ddn-20231215-3
[32] Retrieved from: https://digital-strategy.ec.europa.eu/en/policies/ai-factories

Feature	Description	Level of importance
Regulatory landscape	Government policies and legal frameworks, such as data privacy laws and AI governance models, shape the responsible deployment of AI. Regulations provide guidelines for (ethical) AI development, increasing public and corporate trust in AI.	High
AI Models that address ethical and transparency concerns	Many AI models operate as "black boxes," making their decision-making processes difficult to interpret. Ethical risks include AI bias, discrimination, and lack of cultural adaptability. AI systems sometimes reinforce existing inequalities due to bias in data. The use of AI in sensitive areas (e.g., criminal justice, hiring, or surveillance) raises human rights and accountability concerns. The availability of AI Models that address ethical and transparency concerns can increase trust in the developed and adopted AI.	High
Private investments	Availability and accessibility to private investments (venture capital and private equity) support AI research, development and adoption.	High
Public investments in AI	Public funding is an important driver for AI development and uptake at the EU, national and regional levels.	High
Open-source data and models	Open-source AI frameworks and datasets accelerate research and democratise access to the most advanced models. Transparency through open-source also improves model reliability, security, and ethical considerations, boosting trust in AI solutions.	Medium
Diversity in research areas out of AI	Active support of research on classical (non-neural) approaches in AI, as well as research that combines neural approaches with other approaches, (classical AI) and that integrates various paradigms into more complete cognitive architectures drive innovation in AI and its adoption.[33]	Medium
AI literacy (b**asic understanding and benefits** of AI)	Greater public awareness and AI literacy drive higher adoption rates across industries and daily life. Effectively communicating AI's benefits, such as automation, efficiency, and enhanced decision-making, while also addressing potential risks fosters broader acceptance and integration.	Medium

[33]AAAI 2025 Presidential Panel on the Future of AI Research. https://aaai.org/wp-content/uploads/2025/03/AAAI-2025-PresPanel-Report-FINAL.pdf

Table 4 - Overview of the drivers for the development and uptake of AI. Source: Technopolis Group (2025)

2.4.2 Barriers to the development and uptake of AI

Feature	Description	Level of importance
Barriers related to AI technologies generation and adoption in Europe		
Lack of AI computing capacity	Considering the data centre/cloud capacity gap there is a need for European data centre capacity to triple by 2027 and reach around 22.700 MW[34]. Such a gap has significant implications for Europe's opportunities to pursue technological sovereignty and competitiveness, as well as its ability to boost AI development.	High
Lack of (high-quality) data	AI relies on high-quality datasets; however, data is often fragmented, unavailable, or difficult to integrate. Privacy laws, competitive concerns, and lack of interoperability hinder data access. Many businesses and organisations still rely on legacy systems that do not support AI-friendly data structures.	High
Costs of construction and equipment for AI	Specialised AI equipment such as AI-optimised GPUs is more costly than previous generation equipment, increasing costs for investments in AI uptake.	High
Private investments (venture capital and private equity) **in AI in the EU are lagging**	According to the European Court of Auditors, despite the EU having a strong AI public research community (the highest number of peer-reviewed scientific publications on AI in the world in 2022), it faces challenges in the global race for AI investment. Private investment in AI has been lower than in other AI-leading regions of the world (the US and China) since 2015[35].	High
Unclear business case and ROI for internal **investments** in AI	Organisations struggle to justify AI investments due to unclear return on investment (ROI). AI projects often require significant upfront investment with long-term payoffs, making decision-makers hesitant. AI's potential benefits are sometimes overestimated, leading to unrealistic expectations and failed implementations.	High

[34] Savills News. (2024). European data centre power capacity projected to rise to approximately 13,100 MW by 2027. Available at: https://www.savills.be/research-and-news/savills-news/362723/european-data-centre-power-capacity-projected-to-rise-to-approximately-13-100-mw-by-2027

[35] Retrieved from: https://www.eca.europa.eu/ECAPublications/SR-2024-08/SR-2024-08_EN.pdf

Feature	Description	Level of importance
Fragmentation of coordinated initiatives/fundings/incentives along the AI value chain	Fragmentation between various EU actions as well as limited synergies between national and EU actions could slow progress in the take-up and development of AI	High
Lack of incentives for data sharing and experimentation with data	Concerns over liability and data ownership prevent widespread data sharing to enable advancements in AI. Companies hesitate to share data due to privacy concerns, competitive risks, and legal uncertainties. There is a lack of widely accepted standardised frameworks for secure and fair data exchange. Public institutions often hold valuable datasets but have limited mechanisms to share them with businesses and researchers.	High
Limited access to technology infrastructures	AI development requires access to advanced infrastructure, but many SMEs and startups lack access to these resources. High-performance computing, AI testbeds, and cloud platforms are costly and not equally available across the EU.	High
Complex regulatory landscape	AI providers are subject to the regulatory framework that concerns the data used for AI development and use, affecting cloud and AI providers. This includes the GDPR, the AI Act, DMA, the Data Act, the Free Flow of non-personal data regulation and others adopted in the wake of the EU Data Strategy. As indicated in the Draghi report, the regulatory barriers to scaling in the technology sector are particularly burdensome, especially for startups and the EU's regulatory approach stifles innovation, with approximately 100 tech-focused laws and over 270 regulators overseeing digital networks across all Member States[36]. This also includes data privacy constraints in AI training. Additionally, the complex regulatory landscape is related to a limited risk approach in policy-making that can further hinder AI development and uptake.	High
Speed of AI development	Increasing number of AI models creates challenges in navigating the landscape and making the right choice of the most suitable model for organisations. More specifically, choosing the optimal algorithm necessitates a thorough understanding of the trade-offs involved, such as balancing between the algorithm's accuracy and its interpretability, computational efficiency, and suitability for	Medium

[36] Retrieved from: https://commission.europa.eu/topics/eu-competitiveness/draghi-report_en

Feature	Description	Level of importance
	real-time operations. While deep learning methods may deliver unparalleled accuracy, their often-opaque nature may not be ideal for industries where explainability is crucial for safety and compliance reasons.[37]	
Incompatibility with existing infrastructures	A significant portion of the industry relies on legacy systems, including Enterprise Resource Planning (ERP), Customer Relationship Management (CRM) systems, Product Lifecycle Management (PLM), Manufacturing Execution System (MES), and Laboratory Information Management Systems (LIMS). These platforms were often not designed to accommodate the integration of new AI technologies[38]. Moreover, the processing requirements for AI, particularly with advanced deep learning models are substantial. Industrial infrastructure, however, may not always be up to date or able to support these demands. Upgrading hardware and systems to support the computational intensity of AI can be a daunting and expensive obstacle for many organisations.[39]	Medium
Trust in models and outcomes	Lack of trust in models and their outcomes serves as a significant barrier to AI adoption. When users and stakeholders are uncertain about how a model makes decisions, they may hesitate to rely on its outputs, fearing inaccuracies, biases, or unintended consequences. A lack of transparency exacerbates this issue.[40]	Medium
Fear or AI replacing workers hindering adoption (e.g. certain health domains)	Routine and repetitive tasks are the ones AI is most likely to automate, according to the World Economic Forum's Jobs of Tomorrow whitepaper[41], whereas critical thinking and complex problem-solving could be augmented by the technology. In addition, in the next five years, employees estimate that 44% of workers' skills will be disrupted, meaning upskilling and life-long learning are now more essential, according to the Future of Jobs Report 2023.[42] Consequently,	Medium

[37] Sinha, S., & Lee, Y. M. (2024). Challenges with developing and deploying AI models and applications in industrial systems. Discover Artificial Intelligence, 4(1), 55.
[38] Sinha, S., & Lee, Y. M. (2024). Challenges with developing and deploying AI models and applications in industrial systems. Discover Artificial Intelligence, 4(1), 55.
[39] Sinha, S., & Lee, Y. M. (2024). Challenges with developing and deploying AI models and applications in industrial systems. Discover Artificial Intelligence, 4(1), 55
[40] Retrieved from: https://www.mckinsey.com/capabilities/quantumblack/our-insights/building-ai-trust-the-key-role-of-explainability
[41] Retrieved from: https://www.weforum.org/stories/2023/12/ai-fobo-jobs-anxiety/
[42] Retrieved from: https://www.weforum.org/stories/2023/12/ai-fobo-jobs-anxiety/

Feature	Description	Level of importance
	according to McKinsey report, about half of employees worry about AI inaccuracy and cybersecurity risks[43].	
Environmental impact and carbon footprint of AI	AI models, especially deep learning, consume vast computational resources, increasing energy use and carbon emissions. The rapid scaling of AI requires significant amounts of electricity, often powered by non-renewable sources. AI data centres contribute to global energy consumption, raising sustainability concerns. There is a lack of regulatory measures to ensure AI development follows sustainable practices.	Medium

Table 5 - Overview of barriers to AI technologies generation and adoption in Europe. Source: Technopolis Group (2025)

[43] Retrieved from: https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/superagency-in-the-workplace-empowering-people-to-unlock-ais-full-potential-at-work

2.5. Emerging technologies

The emerging technologies related to artificial intelligence and machine learning show a strong emphasis on addressing challenges related to data privacy, security, and explainability. According to a recent report by JRC[44] and to AAAI[45] and data collected for this study, common novel aspects of AI are included in the figure below.



Figure 12 - Emerging AI technologies. Source: Technopolis Group (2025).

More specifically:

- **AI for scientific discovery:** Artificial Intelligence is revolutionising scientific discovery by accelerating the entire research cycle from knowledge extraction and hypothesis generation to automation of experimentation and verification at an unprecedented speed.
- **Artificial general intelligence (AGI**): Unlike narrow AI, which is designed for specific tasks (e.g., language translation, image recognition), AGI aspires to exhibit human-like cognitive abilities. In a broad sense, AGI is capable of understanding and responding to any kind of problem. In contrast AI agents are designed for designed for specific tasks like chatbots or recommendation systems.
- **Decentralised and federated approaches**: Technologies such as Federated Machine Learning, Federated Deep Learning, and Vertical Federated Learning, adopt decentralised

[44] European Commission: Joint Research Centre, Eulaerts, O., Grabowska, M. and Bergamini, M., Weak signals in science and technologies 2024 – Technologies at an early stage of development that could impact our future, Publications Office of the European Union, 2025, https://data.europa.eu/doi/10.2760/6571994 &
[45]AAAI 2025 Presidential Panel on the Future of AI Research. https://aaai.org/wp-content/uploads/2025/03/AAAI-2025-PresPanel-Report-FINAL.pdf

and federated approaches to enable collaborative model training while preserving data privacy and security.

- **AI agents:** Agents and multi-agent systems (MAS) have evolved from autonomous problem-solving entities to integrating generative AI and LLMs, ultimately leading to cooperative AI frameworks that enhance adaptability, scalability, and collaboration.
- **Emphasis on human-AI collaboration and ethics**: Human-Centric AI, Ethical AI, and Human AI Interface technologies prioritise human-AI collaboration, ethics, and inclusivity, striving for more accountable, fair, and transparent AI systems.
- **AI factuality & trustworthiness:** Improving factually and trustworthiness of AI systems is currently one of the largest topics of AI research.
- **Advanced cryptographic methods and secure multi-party computation**: Privacy-preserving machine learning and other technologies leverage advanced cryptographic methods and secure multi-party computation to ensure data security and regulatory compliance during model training and inference.
- **Adaptability to analyse heterogeneous data and environments:** Technologies like Multimodal AI, Masked Face Recognition, and Tiny Machine Learning demonstrate adaptability to diverse data types, environments, and edge devices, enabling more comprehensive and nuanced understanding in complex domains.
- **AI evaluation:** AI evaluation is the process of assessing the performance, reliability, and safety of AI systems.
- **Embodied AI:** Embodied AI creates intelligent agents that perceive, understand, and interact with the physical world.
- **Hardware & AI:** Hardware/software architecture co-design for artificial intelligence involves creating hardware and software components that are specifically designed to work together efficiently, maximising the performance and energy efficiency of AI systems**.**
- **AI & sustainability:** AI is rapidly transforming industries and holds immense potential to drive sustainability progress, ranging from accelerating the net-zero energy transition to enhancing climate resilience. However, its deployment also raises challenges, such as increasing energy and water demands. Ensuring AI advances sustainability rather than exacerbating environmental risks will require proactive efforts to shape its development, operations, and applications.

2.6. Cross-technological synergies

In terms of cross-technological synergies between AI and other technologies, some examples of such synergies are provided in a table below. It is also important to note that while AI can be an enabler in many other technology fields (“AI for X”), but AI can also benefit from progress in some of these fields (“X for AI”), for instance through the development of the so-called AI chips.

Technology area	Description
Biotechnology	AI-Biotech synergies largely concern AI-enhanced biological processes, fastened drug development, logistic and supply chains, industry data

	management, early diseases identification.[46] Biological understanding of intelligence can also yield insights into AI, as neural networks showed, but also in neuromorphic chips (inspired by the human brain) for example.
High Performance Computing (HPC)	HPC and AI can be used in a variety of fields, including researchers working in laboratories, engineers solving complex technical problems.[47] HPC provides the computational capacity needed to process vast amounts of data.
Microelectronics	Artificial Intelligence can reduce the time and cost of designing new wireless chips and discovering new functionalities to meet expanding demands for better wireless speed and performance[48]
Robotics, including drones, autonomous ground vehicles	Embedded AI refers to AI integrated directly into robotic systems, enabling real-time decision-making, automation, and adaptability without relying on cloud computing. For instance, AI-driven NLP allows robots to understand and respond to human commands.
Quantum Computing	Quantum Computing can enable AI to process larger data volumes efficiently, identify complex patterns better than traditional AI, and improve model performance. The rise of Quantum AI can also enhance security, communication and system resilience.[49]
Big Data	Big Data forms the backbone of modern AI and Data Science, fuelling technologies like Machine Learning, Data Mining, and cloud computing by providing the data necessary for training algorithms and making predictions. Big Data refers to the vast amounts of structured, semi-structured and unstructured data generated at high speeds from various sources, such as social media platforms, e-commerce transactions, and devices connected to the Internet of Things. Traditional data processing techniques are not capable of handling this enormous and fast-moving data. The concept of Big Data is often defined by the "three Vs": Volume (the sheer amount of data), Variety (the diverse types of data), and Velocity (the speed at which data is generated and processed).
Digital Security & Cybersecurity	Cybersecurity organisations increasingly rely on AI in conjunction with more traditional tools such as antivirus protection, data-loss prevention, fraud detection, identity and access management, intrusion detection, risk management and other core security areas. Because of the nature of AI, which can analyse enormous sets of data and find patterns, AI is uniquely suited to tasks such as detecting actual attacks more accurately than humans, creating fewer false-positive results, and prioritising responses based on their real-world risks[50] Additionally, AI has the potential to be a game-changing tool in penetration testing, intentionally probing the defences of software and networks to identify weaknesses.

[46] Retrieved from: https://www.nato.int/nato_static_fl2014/assets/pdf/2023/3/pdf/stt23-vol2.pdf
[47] Retrieved from: https://www.nvidia.com/en-us/high-performance-computing/hpc-and-ai/
[48] Retrieved from: https://engineering.princeton.edu/news/2025/01/06/ai-slashes-cost-and-time-chip-design-not-all
[49] Retrieved from: https://www.nato.int/nato_static_fl2014/assets/pdf/2023/3/pdf/stt23-vol2.pdf
[50] Retrieved from: https://www.morganstanley.com/articles/ai-cybersecurity-new-era

Digital Twins	AI can structure inputs and synthesise outputs of digital twins, and digital twins can provide a robust test-and-learn environment for AI. For example, LLMs could potentially create code[51] for the digital twin, accelerating the development process and increasing effectiveness. Also, multimodal LLMs can efficiently analyse and interpret large data sets and synthesise insights. Additionally, LLMs can generate scenarios for the digital twin to simulate that are more comprehensive. In this way, LLMs (such as OptiGuide or Vertex) can function as an interface for simulators, allowing users to communicate with digital twins via natural language (for example, asking questions and receiving understandable insights in return)[52].

Table 6- Examples of Cross-technological synergies of AI with other technologies. Source: Technopolis Group (2025)

2.7. Global landscape & market dynamics

The main market and global trends related to AI are described below:

- **AI research:** Between 2010 and 2022, the total number of AI publications nearly tripled, rising from approximately 88,000 in 2010 to more than 240,000 in 2022. The increase over the last year was a modest 1.1%.[53]
- **Technology generation**: From 2021 to 2022, AI patent grants worldwide increased sharply by 62.7%. Since 2010, the number of granted AI patents has increased more than 31 times[54]. There is a significant gap between the EU's share of AI patents and the share held by China and the US – in fact, the EU's share of AI patents is about half that of the US[55].
- **Technology adoption and investment**: In the EU, there is a discrepancy between the uptake of and investments in AI. On the one hand, there is a significant increase in the adoption rate of AI. At the same time, 59% of businesses reported investing less than 1% of their annual turnover in AI[56]. Moreover, AI is still evolving, and investing in the development of proprietary AI systems is often seen as risky, particularly for SMEs.
- **Private investments (venture capital and private equity)**: The US is leading private investment in AI (€62.5 billion) in 2023, followed by China (€7.3 billion). The EU and the United Kingdom (UK) together attracted €9 billion worth of private investment in 2023[57]. In terms of VC & PE attracted by companies, the United States lead with OpenAI's generative AI, Waymo's self-driving product, and xAI. In China, Moonshot AI dominates with the chatbot

[51] Nevertheless, it should be noted that to date, research from GitClear based on analysis of 211 million lines of code suggests that AI coding assistants are eroding code quality by increasing duplicated and copy/pasted code and decreasing refactoring. Consequently, for future application of AI to create code with other technologies, the limited quality of the generated code must be addressed.
Source: https://devclass.com/2025/02/20/ai-is-eroding-code-quality-states-new-in-depth-report/
[52] Retrieved from: https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/tech-forward/digital-twins-and-generative-ai-a-powerful-pairing
[53] Retrieved from: https://www.mediahuis.ie/app/uploads/2024/04/HAI_AI-Index-Report-2024.pdf
[54] Retrieved from: https://www.mediahuis.ie/app/uploads/2024/04/HAI_AI-Index-Report-2024.pdf
[55] Commission Staff Working Document - *Second annual report on key findings from the European Monitor of Industrial Ecosystems (EMI)* Accompanying the document Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions The 2025 Annual Single Market and Competitiveness Report (2025)
Retrieved from: https://single-market-economy.ec.europa.eu/document/download/f8d5d980-eba4-4fd7-90ad-0d01c3e2855f_en?filename=SWD_2025_12_F1_OTHER_STAFF_WORKING_PAPER_EN_V5_P1_3878248.PDF
[56] Ibid.
[57] *Ibid.*

Kimi. In Europe, Mistral AI, is redefining efficiency and scalability in AI development and to push the boundaries of Open-Source AI[58].

- **Impact on productivity and competitiveness:** A McKinsey survey reveals that 59% of surveyed organisations report revenue increases from implementing AI (including generative AI)[59].
- **Skills**: in 2024, based on Lightcast data, Machine Learning represents 22% of AI skill demand at the EU level. Meanwhile, cutting-edge technologies like Deep Learning and Generative AI are increasingly revolutionising hiring trends, signalling the beginning of a new era in workforce development[60].

Specific technological developments:

- In terms of **Generative Artificial Intelligence (GenAI)**, the value chain is largely dominated by a few major US players, while China is in the race for global leadership and holds a dominant position[61]. Europe's technology sovereignty as well as its economic and geopolitical development and leadership are at stake GenAI disruption is also an opportunity for Europe to provide alternatives to foreign technologies.[62]
- In 2023, a total of 149 **foundation models** were released, more than double the amount released in 2022. Of these newly released models, 65.7% were open-source, compared to only 44.4% in 2022 and 33.3% in 2021.In 2023, 61 AI models originated from U.S.-based institutions, far outpacing the European Union's 21 and China's 15[63]. However, on 10 select AI benchmarks, closed models outperformed open ones, with a median performance advantage of 24.2%.

[58] Retrieved from: https://aiworld.eu/
[59] Retrieved from: https://www.mediahuis.ie/app/uploads/2024/04/HAI_AI-Index-Report-2024.pdf
[60] Retrieved from: https://aiworld.eu/
[61] Retrieved from: https://single-market-economy.ec.europa.eu/document/download/f8d5d980-eba4-4fd7-90ad-0d01c3e2855f_en?filename=SWD_2025_12_F1_OTHER_STAFF_WORKING_PAPER_EN_V5_P1_3878248.PDF
[62] Retrieved from: https://single-market-economy.ec.europa.eu/document/download/f8d5d980-eba4-4fd7-90ad-0d01c3e2855f_en?filename=SWD_2025_12_F1_OTHER_STAFF_WORKING_PAPER_EN_V5_P1_3878248.PDF
[63] Retrieved from: https://www.mediahuis.ie/app/uploads/2024/04/HAI_AI-Index-Report-2024.pdf

3. A proposed Taxonomy on AI Technology

3.1. Approach towards taxonomy building

To develop a taxonomy of AI, multiple sources of information were analysed, including: 1) foresight literature reports; 2) portfolios of leading RTOs and research institutes; 3) Strategic Research and Innovation agendas of European Partnerships; 4) ERC and EIC project portfolio; 5) Existing taxonomies.

The table below provides an overview of recent literature sources providing an AI taxonomy.

• OECD AI policy observatory[64]
• Adra (2024) - Policy paper and technology roadmap GenAI and robotics 4EU[65]
• EU-U.S. Terminology and Taxonomy for Artificial Intelligence - Second Edition[66]
• JRC (2024). Eyes on the Future - Signals from recent reports on emerging technologies and breakthrough innovations to support European Innovation Council strategic intelligence - Volume 1.2[67]
• AIWorld classification[68]
• AI Index Report (2024): Annual Report. AI Index Steering Committee, Institute for Human-Cantered AI. Authors: Nestor Maslej, Loredana Fattorini, Raymond Perrault et al., Stanford University CA, April 2024.
• Epoch AI, 'Parameter, Compute and Data Trends in Machine Learning'. Published online at epoch.ai. Retrieved from: 'https://epoch.ai/data/notable-ai-models' [online resource]
• EMI Methodological Report (2024)[69]
• Creation of a Taxonomy for the European AI Ecosystem: A Report of the Cross-KIC Activity[70]
• OECD Framework for the Classification of AI systems[71]
• Artificial Intelligence and the Evolution of AI (Model) Capabilities A Conceptual Primer[72]

[64] Retrieved from: https://oecd.ai/en
[65] Retrieved from: https://adr-association.eu/news/adra-releases-policy-paper-and-technology-roadmap-genai-and-robotics-4eu
[66] Retrieved from: https://digital-strategy.ec.europa.eu/en/library/eu-us-terminology-and-taxonomy-artificial-intelligence-second-edition
[67] Bailey, G., Farinha, J., Mochan, A. and Polvora, A., Eyes on the Future - Signals from recent reports on emerging technologies and breakthrough innovations to support European Innovation Council strategic intelligence - Volume 1, Publications Office of the European Union, Luxembourg, 2024, doi:10.2760/355, JRC137811.
[68] Retrieved from: https://aiworld.eu/
[69] Retrieved from:https://monitor-industrial-ecosystems.ec.europa.eu/sites/default/files/2023-12/EMI%20Methodological%20Report.pdf
[70] Retrieved from: https://eit.europa.eu/sites/default/files/creation_of_a_taxonomy_for_the_european_ai_ecosystem_final.pdf
[71] Retrieved from:https://www.oecd.org/en/publications/oecd-framework-for-the-classification-of-ai-systems_cb6d9eca-en.html
[72] Retrieved from: https://media.licdn.com/dms/document/media/v2/D4D1FAQEKkwTxKK78WQ/feedshare-document-pdf-analyzed/B4DZTapqukGkAY-/0/1738835159295?e=1742428800&v=beta&t=oFEg_BxU5XwPthNcbSTap5Z0P8In1JeJDoDWUZL6paE

• Beckert, B., & Kroll, H. (2024). Definition of the research and innovation field of" Artificial Intelligence" and approaches to determining quality (No. 88). Fraunhofer ISI Discussion Papers-Innovation Systems and Policy Analysis.
• CSET (2022): Counting AI Research: Exploring AI Research Output in English- and Chinese-Language Sources. Center for Security and Emerging Technology (CSET). Author: Daniel Chou, Washington D.C.; CSET, July, https://cset.georgetown.edu/publication/counting-ai-research/.
• OECD (2023): What technologies are at the core of AI? An exploration based on patent data. Authors: Flavio Calvino, Chiara Criscuolo, Hélène Dernis, Lea Samek, OECD Artificial Intelligence Papers, No. 6, OECD Publishing, Paris, https://doi.org/10.1787/32406765-en.

Table 7 - Review of recent AI taxonomies

3.2. Proposed AI taxonomy

For this work, the taxonomy was clustered into “innovation in AI”, “innovation with AI” (capabilities) and “AI application areas”. The figure below provides a high-level overview of the proposed AI taxonomy.

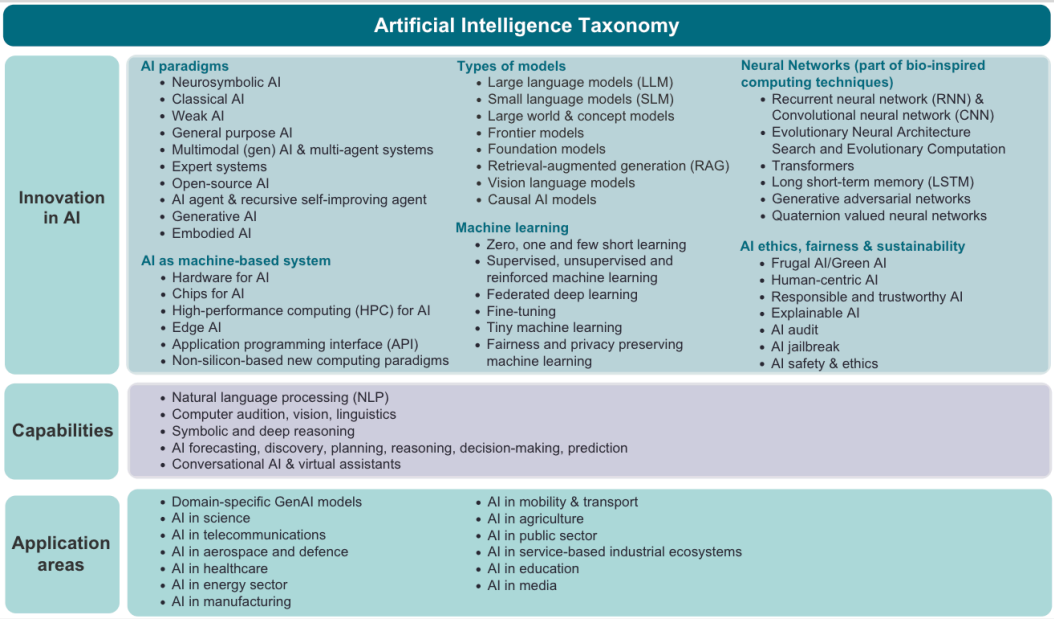


Figure 13 - High-level overview of the proposed AI taxonomy. Source: Technopolis Group (2025)

The table below provides an overview of how our proposed taxonomy is aligned with the guidelines on AI system definition[73].

[73] Retrieved from: https://digital-strategy.ec.europa.eu/en/library/commission-publishes-guidelines-ai-system-definition-facilitate-first-ai-acts-rules-application

EU Guideline on AI system definition	Definition	Proposed taxonomy and argumentation
Machine Based System	The entire lifecycle of advanced AI systems relies on **machines that can include many hardware or software components**. The term 'machine-based' covers a wide variety of computational systems (e.g., traditional computing systems, quantum computing systems).	• **AI as Machine Based System** – **this category of the taxonomy encompasses software and hardware components of AI**. More specifically, it includes Application Programming Interface, High-Performance Computing for AI, Edge-AI synergies and AI hardware components such as graphic processing units etc.
Autonomy	AI systems are designed to operate with 'some degree of independence of actions from human involvement and of capabilities to operate without human intervention	• **AI paradigms –** reflect the human-machine interaction and autonomy level of the AI (e.g., general purpose AI (AGI) is a type of highly autonomous artificial intelligence (AI) that matches or surpasses human cognitive capabilities). • **AI capabilities** – also reflect the capacity of AI to autonomously generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments). • **AI Ethics & Sustainability Fairness** – covers such as trustworthy AI, human-centric AI and explainable AI that are crucial components for AI autonomous operation.
Adaptiveness (self-learning capabilities)	Adaptiveness' refers to self-learning capabilities, allowing the behaviour of the system to change while in use. The new behaviour of the adapted system may produce different results from the previous system for the same inputs.	• **Machine learning** – reflects self-learning approaches (e.g., Self-supervised learning under machine learning) that enable the adaptiveness of AI. • **AI paradigms –** reflect the human-machine interaction (e.g., Agentic AI) and types of AI that can adapt.
AI system objectives (explicit and implicit objectives)	Explicit objectives refer to clearly stated goals that are directly encoded by the developer into the system. Implicit objectives refer to goals that are not explicitly stated but may be deduced from the behaviour or underlying assumptions of the system	• **AI paradigms –** Narrow AI (encompasses artificial intelligence systems purpose-built for specific, well-defined tasks or applications), Agentic AI (a probabilistic technology with high adaptability to changing environments and events).

Inferencing how to generate outputs using AI techniques	Capability to infer refers to the process of obtaining the outputs, such as predictions, content, recommendations, or decisions, which can influence physical and virtual environments, and to a capability of AI systems to derive models or algorithms, or both, from inputs or data. The techniques that enable inference while building an AI system include machine learning approaches that learn from data how to achieve certain objectives, and logic- and knowledge-based approaches that infer from encoded knowledge or symbolic representation of the task to be solved.	• **Machine learning –** inference is the process that a trained machine learning model uses to draw conclusions from brand-new data. An AI model capable of making inferences can do so without examples of the desired result. • **Neural Networks -** Neural networks have become one of the most important tools for artificial intelligence (AI). Well-designed and trained NNs can perform inference (e.g., make decisions or predictions) on unseen inputs with high accuracy. • **Types of models -** With LLMs, for example, the model can infer what word comes next from a prompt and produce sentences and paragraphs.
Outputs that can influence physical or virtual environments	The ability of a system to generate outputs, such as predictions, content, and recommendations, based on inputs it receives and using machine learning and logic and knowledge-based approaches, is fundamental to what AI systems do and what distinguishes those systems from other forms of software. The capacity to generate outputs and the type of output the system can generate is central to understanding the functionality and impact of an AI system.	• **AI capabilities** – reflect capabilities such as NLP, Computer audition, Computer vision, Computer linguistics, Reasoning & decision-making, AI forecasting and prediction, AI (scientific) discovery, AI planning, AI creation / generation, Conversational AI & virtual assistants.
Interaction with the environment	That element should be understood to emphasise the fact that AI systems are not passive, but actively impact the environments in which they are deployed. Reference to 'physical or virtual environments' indicates that the influence of an AI system may be both to tangible, physical objects (e.g. robot arm) and to virtual environments, including digital spaces, data flows, and software ecosystems.	• **AI applications -** capability of AI, i.e. interaction with other systems (both physical such as vehicles or robots or digital such as Digital Twins.

Table 8 - Overview of the alignment between the proposed taxonomy and the Guideline on AI system definition. Source: Technopolis group (2025)

AI applications and AI Ethics, fairness & responsibility categories are cross-cutting and can be attributed to all seven components of AI system definition.

3.2.1. Taxonomy – innovation in AI

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
AI paradigms				
Neuro-symbolic AI (former Hybrid AI)	Neurosymbolic AI (formerly known as Hybrid AI) is an advanced approach to artificial intelligence that combines symbolic reasoning (traditional rule-based AI) with neural networks (machine learning and deep learning). This fusion aims to overcome the limitations of both paradigms, creating AI systems that are more powerful, interpretable, and generalisable.	High	Emerging	https://arxiv.org/abs/2305.00813
Open-source AI	Open-source AI refers to artificial intelligence projects, models, and tools that are publicly available for use, modification, and distribution. These projects are often developed collaboratively by communities of researchers, developers, and organisations. For example: EUROLLM.	High	Emerging	https://interoperable-europe.ec.europa.eu/collection/open-source-observatory-osor/news/eurollm-pioneering-european-open-source-ai
AI Agent	AI Agents represent the integration of multiple AI capabilities, such as reasoning, perception, and action, into a single, autonomous entity capable of performing complex tasks without direct human intervention.	High	R&D	https://ai-world.eu/technology/ai-agent

[74] Discussed during the validation workshop
[75] Discussed during the validation workshop

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
	These AI Agents are more sophisticated than simple algorithms or reactive systems, blending different technologies like Reinforcement Learning, NLP, and Computer Vision. Historically, the concept of AI Agents emerged as researchers sought to build systems that could not only make decisions but also adapt, learn from experience, and interact with their environment dynamically. This has allowed AI Agents to transition from basic virtual assistants to systems that power autonomous vehicles, robotic machinery, and advanced recommendation systems.			
Multi-agent systems	A multi-agent system (MAS or "self-organised system") is a computerised system composed of multiple interacting intelligent agents.[1] Multi-agent systems can solve problems that are difficult or impossible for an individual agent or a monolithic system to solve. MAS intelligence may include methodic, functional, procedural approaches, algorithmic search or reinforcement learning. With advancements in LLMs, LLM-based multi-agent systems have emerged as a new area of research, enabling more sophisticated interactions and coordination among agents. Unlike single-agent systems that operate independently, multi-agent systems orchestrate multiple AI agents to work together toward shared goals.	High	R&D	https://techcommunity.microsoft.com/blog/aiplatformblog/the-future-of-ai-exploring-multi-agent-ai-systems/4226593
Generative AI	Generative AI (GenAI) refers to a branch of Artificial Intelligence that goes beyond traditional AI tasks like classification and prediction by creating entirely new outputs. Unlike models that simply identify or categorise patterns, GenAI models, such as	High	Deployment	https://ai-world.eu/concept/generative-ai

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
	Generative Adversarial Networks and Transformer-based models like GPT, learn to replicate patterns in data and generate novel content, whether it be text, images, music, or code. At its core, GenAI aims to mimic human creativity, enabling machines to generate content that resembles human-generated work. Built upon Deep Learning models, GenAI technologies are transforming AI from purely functional tools into creators capable of producing original content that can closely resemble human creativity.			
Recursive Self-improving Agents	Self-improvement can happen at multiple levels of sophistication. Narrow self-improvement: In this category, an agent improves its performance within the context of a fixed operating environment or goal. A simple form of such narrow improvement is an LLM-based agent that monitors its own performance and can autonomously launch a fine-tuning loop to retrain its LLM on a new dataset when it detects that its performance is deviating from an acceptable range. Broad self-improvement: This category encompasses broader and more sophisticated modalities including agents that can create tools, modify their own architecture, and even create new agents	Medium	Emerging	https://www.emergence.ai/blog/self-improving-agents Validation Workshop
Decision intelligence	Decision intelligence is the application of AI and machine learning technologies, along with data fusion, data visualisation and collaboration tools, to augment and improve decision making.	Medium	Emerging	https://www.cognyte.com/nexyte/what-is-decision-intelligence/ Validation Workshop

Innovation in AI				
Keyword	**Description**	**Disruption potential**[74]	**Technology maturity**[75]	**Source**
Multi-modal (gen) AI	Multimodal Generative AI refers to artificial intelligence models that can process and generate multiple types of data, such as text, images, audio, and video, simultaneously. Unlike unimodal AI, which specialises in a single type of data (e.g., text-only models like GPT), multimodal AI integrates and understands different data formats to produce richer and more context-aware outputs.	Medium	R&D	https://www.ibm.com/think/topics/multimodal-ai
Embodied AI	Embodied AI creates intelligent agents that perceive, understand, and interact with the physical world.	Medium	R&D	Workshop
Weak AI/Narrow AI/Specific AI	Narrow AI, also known as Weak AI or Artificial Narrow Intelligence, refers to AI systems that are designed to perform a specific task or a set of closely related tasks.	Medium	R&D	https://deepai.org/machine-learning-glossary-and-terms/narrow-ai
General purpose AI/Strong AI	Strong artificial intelligence, or strong AI, also known as artificial general intelligence (AGI) or general AI, is a hypothetical form of AI that, if it could be developed, would possess intelligence and self-awareness equal to those of humans, and the ability to solve an unlimited range of problems.	Medium	R&D	https://www.ibm.com/think/topics/strong-ai
Classic Symbolic AI	Symbolic AI uses tools such as logic programming, production rules, semantic nets and frames, and it developed applications such as knowledge-based systems (in particular, expert systems), symbolic mathematics, automated theorem provers, ontologies, the semantic web, and automated planning and scheduling systems.	Medium	Deployment	https://knowledge-zone.co.in/posts/6592594431e96c930978f096 Workshop

Innovation in AI				
Keyword	**Description**	**Disruption potential**[74]	**Technology maturity**[75]	**Source**
Expert system	An expert system is a computer programme that uses artificial intelligence (AI) technologies to simulate the judgment and behaviour of a human or an organisation that has expertise and experience in a particular field. Expert systems are usually intended to complement, not replace, human experts.	Low	Deployment	https://onlinelibrary.wiley.com/journal/14680394
Types of models				
Vision Language Models	Vision language models are models that can learn simultaneously from images and texts to tackle many tasks, from visual question answering to image captioning	High	R&D	https://huggingface.co/blog/vlms
Foundation models	Foundation models are artificial intelligence (AI) models trained on vast, immense datasets and can fulfil a broad range of general tasks. They serve as the base or building blocks for crafting more specialised applications.	High	R&D	https://www.ibm.com/think/topics/foundation-models
Small Language Models (SLM)	Small language models (SLMs) are compact versions of LLMs designed to deliver efficient NLP capabilities while requiring significantly fewer computational resources. Unlike massive models with billions of parameters, SLMs typically have a few million to a few hundred million parameters.[76]	High	Deployment	https://www.ibm.com/think/topics/small-language-models
Large language models	LLMs are a category of foundation models trained on immense amounts of data making them capable of understanding and generating natural language and other types of content to perform a wide range of tasks.	High	Deployment	https://www.ibm.com/think/topics/large-language-models

[76] In terms of size, SLM parameters range from a few million to a few billion, as opposed to LLMs with hundreds of billions or even trillions of parameters. Parameters are internal variables, such as weights and biases, that a model learns during training

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
Frontier Models	Frontier Models represent the cutting-edge of Artificial Intelligence, referring to the most advanced and powerful AI systems pushing the boundaries of what is technologically possible. These models are typically large-scale Neural Networks with immense computational power, often designed to tackle unprecedented challenges in fields such as NLP, and multi-modal tasks. Frontier Models aim to solve increasingly complex problems that traditional models struggle with, from strategic decision-making in games to real-time language understanding across diverse contexts. These models are frequently associated with high-impact AI research, marking significant milestones in the development of more sophisticated and capable AI systems.	Medium	Emerging	https://ai-world.eu/concept/frontier-models
Large world model (LWMs)	LWMs may represent the future of AI, extending beyond text, audio and images to include the entire spectrum of our physical and digital realities. LWMs will process real-world data from various sources, such as IoT devices, sensors, cameras and more, to comprehend and interact with the world in a way that mirrors human perception and cognition. World Models (WMs) will excel in processing diverse data inputs and surpass traditional limitations.	Medium	R&D	https://www.forbes.com/councils/forbestechcouncil/2024/01/23/the-next-leap-in-ai-from-large-language-models-to-large-world-models/ Workshop
Large Concept Models LCMs	Concepts (architecture which operates on an explicit higher-level semantic representation) are language- and modality-agnostic and represent a higher-level idea or action in a flow. The Large Concept Model is trained to perform autoregressive sentence prediction in an embedding space.	Medium	R&D	https://ai.meta.com/research/publications/large-concept-models-language-modeling-in-a-sentence-

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
				representa-tion-space/ Workshop
Retrieval Aug-mented Genera-tion	Retrieval Augmented Generation (RAG) is a hybrid approach that combines two core AI techniques: retrieval-based methods and generative models. In RAG systems, a model first retrieves relevant information from external sources or databases and then uses a generative model, like a Transformer, to produce a coherent and contextually appropriate response based on the retrieved data. This approach allows the system to generate more accurate and up-to-date answers by relying on external knowledge, making it particularly useful for tasks that require factual correctness or up-to-date information. By incorporating retrieval mechanisms, RAG enhances the generative model's ability to draw from a broader pool of information rather than relying solely on what it has been pre-trained on.	Medium	Deployment	https://ai-world.eu/technology/
Causal AI models	Causal AI is a form of AI designed to identify and understand the cause and effect of relationships across data	Low	R&D	https://www.tech-target.com/whatis/definition/causal-AI Workshop
Neural Networks (part of bio-inspired computing techniques)				
Trans-formers	Transformers are a type of Neural Network architecture designed to process sequential data, focusing on understanding the relationships and context between elements in a sequence. Originally introduced in the 2017 paper Attention is All You Need by Google, Transformers marked a	High	Deployment	https://ai-world.eu/technology/transformers

Innovation in AI				
Keyword	**Description**	**Disruption potential**[74]	**Technology maturity**[75]	**Source**
	major leap in NLP by enabling models to capture long-range dependencies in data more effectively than previous methods, like Recurrent Neural Networks and Long Short-Term Memory Networks. The introduction of Transformers revolutionised how machines understand and generate language, leading to innovations in a variety of AI fields.			
Recurrent Neural Networks	A recurrent neural network or RNN is a deep neural network trained on sequential or time series data to create a machine learning (ML) model that can make sequential predictions or conclusions based on sequential inputs. An RNN might be used to predict daily flood levels based on past daily flood, tide and meteorological data. But RNNs can also be used to solve ordinal or temporal problems such as language translation, NLP, sentiment analysis, speech recognition and image captioning.	High	Deployment	https://www.ibm.com/think/topics/recurrent-neural-networks
Attention mechanisms in CNN	Attention mechanism has been regarded as an advanced technique to capture long-range feature interactions and to boost the representation capability for convolutional neural networks.	Medium	R&D	https://arxiv.org/abs/2108.08205
Quaternion Valued Neural Network	Quaternion neural networks have recently received an increasing interest due to noticeable improvements over real-valued neural networks on real world tasks such as image, speech and signal processing. The extension of quaternion numbers to neural architectures reached state-of-the-art performances with a reduction of the number of neural parameters.	Medium	R&D	https://link.springer.com/article/10.1007/s10462-019-09752-1
Evolutionary	Deep Neural Networks (DNNs) have achieved great success in many	Medium	R&D	Weak signals in

Innovation in AI				
Keyword	**Description**	**Disruption potential**[74]	**Technology maturity**[75]	**Source**
Neural Architecture Search and Evolutionary Computation	applications. The architectures of DNNs play a crucial role in their performance, which is usually manually designed with rich expertise. However, such a design process is labour intensive because of the trial-and-error process, and not easy to realise due to the rare expertise in practice. Neural Architecture Search (NAS) is a type of technology that can design the architectures automatically. Among different methods to realise NAS, Evolutionary Computation (EC) methods have recently gained much attention and success.			Science and Technologies - 2024
Convolutional Neural Network (CNN)	Convolutional neural networks are distinguished from other neural networks by their superior performance with image, speech or audio signal inputs. They have three main types of layers, which are: Convolutional layer, Pooling layer, Fully connected (FC) layer.	Medium	Deployment	https://www.ibm.com/think/topics/convolutional-neural-networks
Long Short-Term Memory (LSTM)	Long Short-Term Memory (LSTM) networks are a type of Recurrent Neural Network designed to better capture long-term dependencies in sequential data. Unlike traditional RNNs, which struggle with remembering information over long sequences due to the vanishing gradient problem, LSTMs incorporate memory cells that can retain information for long durations. This makes LSTMs especially useful for tasks that require context from earlier in the sequence, such as speech recognition, machine translation, and time series forecasting. Introduced in the 1990s by Hochreiter and Schmidhuber, LSTMs have since become a fundamental architecture in sequence-based Machine Learning tasks.	Medium	Deployment	https://ai-world.eu/technology/long-short-term-memory-network

Innovation in AI				
Keyword	**Description**	**Disruption potential**[74]	**Technology maturity**[75]	**Source**
Generative Adversarial Networks	Generative Adversarial Networks (GANs) are a class of Machine Learning models designed to generate new data samples that resemble a given dataset. Introduced by Ian Goodfellow in 2014, GANs consist of two Neural Networks, a generator and a discriminator, that compete against each other in a zero-sum game. The generator creates fake data, while the discriminator attempts to distinguish between real and fake data. Over time, the generator becomes increasingly better at producing realistic data that can fool the discriminator. GANs are widely used in various fields, from creating realistic images to generating synthetic data for training other AI models, pushing the boundaries of AI's creative and generative capabilities.	Medium	Deployment	https://ai-world.eu/technology/generative-adversarial-networks
AI as Machine Based System				
Quantum for AI and AI for Quantum	AI has opened new avenues for automated control of quantum systems. In particular, the application of AI can provide invaluable insight into the complex and multifaceted domain of quantum physics to accelerate the discovery of quantum physics laws, and can potentially alleviate challenges that have been historically associated with QC and quantum communication. On the other hand, QC can also be used to enhance AI applications	High	Emerging	https://link.springer.com/article/10.1007/s11082-023-04914-6 Workshop
Chips for AI	AI chips include graphics processing units (GPUs), field-programmable gate arrays (FPGAs), and application-specific integrated circuits (ASICs) that are specialised for AI. General-purpose chips like central processing units (CPUs) can also be used for some simpler AI tasks, but	High	R&D	https://cset.georgetown.edu/publication/ai-chips-what-they-are-and-why-they-matter/

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
	CPUs are becoming less and less useful as AI advance			Workshop
HPC for AI and AI for HPC	HPC and AI can be used in a variety of fields, including researchers working in laboratories, engineers solving complex technical problems. More specifically: • Data driven ML algorithms (probabilistic) that generate the model • New workloads need optimisation of HPC systems for AI tasks • New applications on texts and images • With Deep Learning computational power becomes one key element (data - algorithms - compute). Overall, AI can make HPC far more productive and powerful by analysing and optimising HPC simulations, reducing energy consumption and even improving the algorithms used. On the other hand, HPC makes AI possible, by providing the raw power to train extremely large AI models quickly and efficiently. In other words, HPC provide the necessary supercomputing infrastructure. Specifically, HPC enables: • Faster Training Times • Larger Model Scalability • Cost-Effective • Improved Accuracy.	High	R&D	https://www.nvidia.com/en-us/high-performance-computing/hpc-and-ai/ https://www.data-spaces-symposium.eu/wp-content/uploads/2024/03/HPC-meets-the-demands-of-AI-and-Data_consolidated_final.pdf https://www.zettascale.hpc.cam.ac.uk/wp-content/uploads/2023/11/HPC-AI-Primer.pdf https://inldigitallibrary.inl.gov/sites/sti/sti/Sort_66234.pdf
Hardware for	The hardware components refer to the physical elements of the machine, such as processing units, memory, storage devices,	High	Deployment	https://digital-strategy.ec.europa.eu/en/li

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
artificial intelligence	networking units, and input/output interfaces, which provide the infrastructure for computation. More specifically: • Compute performance relies on central processing units (CPUs) and accelerators, graphics-processing units (GPUs), field programmable gate arrays (FPGAs), and application-specific integrated circuits (ASICs). • AI applications have high memory-bandwidth requirements, since computing layers within deep neural networks must pass input data to thousands of cores as quickly as possible. Memory is required, typically dynamic random-access memory (DRAM), to store input data, weight model parameters, and perform other functions during both inference and training. • AI applications generate vast volumes of data, about 80 exabytes per year, which is expected to increase to 845 exabits by 2025. One potential disruption in storage is new forms of non-volatile memory (NVM). New forms of NVM have characteristics that fall between traditional memory, such as DRAM, and traditional storage, such as NAND flash. They can promise higher density than DRAM, better performance than NAND, and better power consumption than both. • AI applications require many servers during training, and the number increases with time. For instance, developers only need one server to build an			brary/commission-publishes-guidelines-ai-system-definition-facilitate-first-ai-acts-rules-application https://www.mckinsey.com/~/media/McKinsey/Industries/Semiconductors/Our%20Insights/Artificial%20intelligence%20hardware%20New%20opportunities%20for%20semiconductor%20companies/Artificial-intelligence-hardware.ashx

Innovation in AI				
Keyword	**Description**	**Disruption potential**[74]	**Technology maturity**[75]	**Source**
	initial AI model and under 100 to improve its structure.			
Non-silicon based new paradigms	Although these paradigms often differ from one another significantly, there are three common characteristics: the techniques are not based on the traditional silicon-based integrated circuit technology; the basic computing elements have been physically implemented; and many of them aim at miniaturisation.	Medium	Emerging	https://www.techrxiv.org/users/710557/articles/1226154/master/file/data/Wet-Neuromorhpic%20Computin_Arxiv/Wet-Neuromorhpic%20Computin_Arxiv.pdf Workshop
Edge AI	The fusion of edge computing and AI involves processing AI algorithms on users' devices, offering benefits like reduced latency, energy efficiency, and real-time applications. This integration allows for real-time data processing and decision-making at the source, significantly declining latency and bandwidth use. The combination of edge computing and AI enables the development of smarter and more responsive applications, such as autonomous vehicles, industrial IoT, smart home systems, etc. By leveraging edge AI, organisations can achieve greater efficiency, enhanced privacy, and faster insights, driving innovation across various sectors.	Medium	R&D	https://www.ibm.com/think/topics/edge-ai https://arxiv.org/abs/2407.04053
Application	An Application Programming Interface (API) is a set of tools, protocols, and standards that allow different software applications to communicate and interact. In a nutshell, APIs	Medium	Deployment	https://ai-world.eu/technology/application-

Innovation in AI				
Keyword	**Description**	**Disruption potential**[74]	**Technology maturity**[75]	**Source**
Programming Interface	act as intermediaries, enabling developers to access services and functionalities from external systems without needing to understand their internal workings. This makes APIs essential for modern software ecosystems, as they foster integration between diverse services and platforms. In the realm of AI, APIs are particularly valuable, as they allow AI Agents to extend their capabilities by connecting with third-party services. Instead of building every function internally, an AI agent can use APIs to gather data, access external Machine Learning models or control external applications. Other benefits of Integrating APIs with AI include: • Enhanced Functionality: APIs allow AI capabilities to be easily integrated into existing applications, adding advanced features without extensive redevelopment. • Scalability: APIs provide a scalable way to access AI services, allowing applications to handle varying loads efficiently. • Interoperability: APIs enable different AI systems and data sources to work together seamlessly, enhancing the overall functionality of the application.			program-ming-inter-face Chintha-patla, Y. (2024). Integrating APIs with AI: Revolutionising Modern Applications.
Machine Learning				
Tiny machine learning	Tiny machine learning is broadly defined as a fast-growing field of machine learning technologies and applications including hardware, algorithms, and software capable of performing on-device sensor data analytics at extremely low power, typically in the mW range and below, and	High	R&D	Weak signals in Science and Technologies - 2024

Innovation in AI				
Keyword	**Description**	**Disruption potential**[74]	**Technology maturity**[75]	**Source**
	hence enabling a variety of always-on use-cases and targeting battery operated devices.			
Privacy-preserving machine learning	Approaches to technically built privacy-by-design into ML. An ML process involves many assets which may need to be protected. These include: • Identity Privacy. Protecting the data contributor's identity is necessary in many data collection processes, especially when the data collected may reveal political beliefs, health conditions etc. Achieving identity privacy can be seen as ensuring anonymity of the data contributors both from the model creators • Raw Dataset Privacy. Often, the raw dataset Z contains sensitive information, such as the X-ray images from the cancer diagnosis example, and if compromised the patient's privacy can be violated. • Privacy of the model refers to the secrecy of the produced model and its parameter. Keeping the model secret is a common goal desired by the model owners, who want to stay ahead of their competitors. Examples of privacy preserving techniques & methods include: • Differential privacy – a privacy method that helps reveal useful information about a dataset without revealing any private	High	R&D	https://www.sciencedirect.com/topics/computer-science/privacy-preserving-machine-learning https://www.alexandra.dk/wp-content/uploads/2020/10/Alexandra-Instituttet-whitepaper-Privacy-Preserving-Machine-Learning-A-Practical-Guide.pdf

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
	information about an individual in the dataset • Homomorphic Encryption (HE) is a cryptographic method that allows computing on encrypted data, so that the decrypted output is identical to the output of the computation on the original unencrypted input. • Multi-Party Computation (MPC) is a technology that allows multiple parties to compute a function on their private inputs without revealing them Private Aggregation of Teacher Ensembles framework (PATE), which trains an ML model with private data and provides strong privacy guarantees.			
Fine-tuning	Fine-tuning is a crucial technique in Machine Learning that allows developers to take a pre-trained model, typically trained on large, generalised datasets, and adapt it for more specific, task-oriented applications. This process, closely associated with Transfer Learning, leverages the knowledge a model has acquired from one domain and applies it to another, more focused task. Fine-tuning is particularly valuable in scenarios where obtaining a large dataset for training from scratch is impractical, making it a popular method in domains like NLP. By fine-tuning a pre-trained model, developers can improve the model's performance on tasks such as sentiment analysis and image classification.	High	Deployment	https://ai-world.eu/technology/fine-tuning
Fairness-aware machine	Approaches to technically built fairness-awareness and bias detection into AI, and specifically ML. These	Medium	R&D	https://link.springer.com/

Innovation in AI				
Keyword	**Description**	**Disruption potential**[74]	**Technology maturity**[75]	**Source**
learning techniques	methods would be helpful for correcting any type of biases. Common fairness metrics include disparate impact, equalised odds, and statistical parity difference. Pre-processing Techniques: These techniques modify the training data to reduce bias before training the model			article/10.1007/s10664-023-10402-y
Zero, One and Few Shot Learning	**Zero shot**: Expands the capabilities of AI models by enabling them to perform tasks they were not explicitly trained on. In this paradigm, models make predictions about classes they have never encountered during the training phase. This is achieved by using auxiliary information or by interpreting relationships between different classes. **One shot**: ridges the gap between few-shot and zero-shot learning, with the model learning from only one example per new class. This modality is crucial in applications like facial recognition systems, where registering a new user with just one photo is desirable. One shot learning often employs sophisticated techniques to extract maximum information from a single example balancing generalisation with specificity. **Few shots**: Addresses scenarios where limited labelled data are available for a new task. In this modality, models learn to generalise from just a few examples, typically between two to five instances per class, an approach that is particularly useful in real-world applications were obtaining large, labelled datasets is impractical or prohibitively expensive. Few-shot learning often leverages meta-learning techniques, whereby the model learns to learn from limited data, enabling rapid adaptation to new tasks.	Medium	R&D	https://media.licdn.com/dms/document/media/v2/D4D1FA-QEKkwTxKK78WQ/feed-share-document-pdf-analyzed/B4DZTapqukGkAY-/0/1738835159295?e=1742428800&v=beta&t=oFEg_BxU5XwPthNcb-STap5Z0P8In1JeJDoD-WUZL6paE

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
Transfer Learning	Transfer Learning is a Machine Learning technique where a model developed for one task is reused as the starting point for a model on a second, related task. Rather than training a model from scratch, Transfer Learning allows the model to leverage knowledge gained from a previous task to improve performance on a new task, especially when the new task has limited data. This technique is particularly powerful in domains where collecting large, labelled datasets is challenging or expensive. Transfer Learning typically involves taking a pre-trained model, modifying its architecture, and fine-tuning it on a new dataset, which accelerates training and improves accuracy.	Medium	Deployment	https://ai-world.eu/concept/transfer-learning
Learning with Reduced Labelling	Weakly supervised labelling setting known as Learning with Reduced Labels with less expensive long-tailed data. Instead of selecting all supervised labels from an extensive sequence of classes, RL only requires annotators to determine whether the limited set of candidate labels contains the correct class label.	Low	R&D	https://arxiv.org/pdf/2403.16469
Federated Deep learning	Federated Learning (FL) is an emerging framework for distributed processing of large data volumes by edge devices subject to limited communication bandwidths, heterogeneity in data distributions and computational resources, as well as privacy considerations.	Low	R&D	Weak signals in Science and Technologies - 2024
Supervised, unsupervised and reinforced	Different types of learning approaches to train algorithms. Supervised learning is a category of machine learning that uses labelled datasets to train algorithms to predict outcomes and recognise patterns.	Low	Deployment	Weak signals in Science and Technologies - 2024

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
machine learning	Unsupervised learning in artificial intelligence is a type of machine learning that learns from data without human supervision. Unlike supervised learning, unsupervised machine learning models are given unlabelled data and allowed to discover patterns and insights without any explicit guidance or instruction. Reinforced learning (RL) is a type of machine learning process that focuses on decision making by autonomous agents. An autonomous agent is any system that can make decisions and act in response to its environment independent of direct instruction by a human user. Robots and self-driving cars are examples of autonomous agents. In reinforcement learning, an autonomous agent learns to perform a task by trial and error in the absence of any guidance from a human user. It particularly addresses sequential decision-making problems in uncertain environments, and shows promise in artificial intelligence development.			
AI Ethics & Fairness & Sustainability				
Human-centric AI	Human-Centric AI is an emerging discipline intent on creating AI systems that amplify and augment rather than displace human abilities. HCAI seeks to preserve human control in a way that ensures artificial intelligence meets our needs while also operating transparently, delivering equitable outcomes, and respecting privacy. According to the AI High-Level Working Group, AI systems need to be human-centric, resting on a commitment to their use in the service of humanity and the common good, intending to improve human welfare and freedom.	High	Emerging	https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-case-for-human-centered-ai https://research.ibm.com/blog/what-is-human-centered-ai

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
Trustworthy AI	Trustworthy AI is an AI system that is reliable, fair, explainable, and secure. It is designed to respect user privacy, uphold transparency, and provide consistent and unbiased outcomes. Trustworthy AI builds confidence among users by adhering to principles such as fairness, accountability, and safety.	High	R&D	https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai
AI Safety	AI safety is a field focused on ensuring that artificial intelligence systems operate as intended, without causing unintended harm or negative consequences. It encompasses the development of principles, techniques, and tools to make AI systems more robust, reliable, and aligned with human values. AI safety is particularly crucial as AI systems grow in capability and autonomy, affecting areas such as decision-making, ethics, and societal impacts. The goal is to mitigate risks ranging from biased algorithms to existential threats posed by superintelligent AI. AI safety involves a diverse set of methodologies aimed at preventing harmful outcomes from AI systems. This includes addressing issues like fairness, accountability, transparency, robustness, and value alignment. In addition to technical approaches, AI safety also considers societal and ethical implications, promoting collaboration between researchers, policymakers, and industry stakeholders. With AI becoming increasingly integrated into critical systems, ensuring its safe deployment is a priority for the technology's long-term benefits.	High	R&D / Deployment	https://ai-world.eu/concept/ai-safety
Explainable AI	Explainable Artificial Intelligence (XAI) is the ability of AI systems to provide clear and understandable explanations for their actions and decisions. Its central goal is to make the	Medium	Emerging	https://www.edps.europa.eu/system/files/2023-11/23-11-

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
	behaviour of these systems understandable to humans by elucidating the underlying mechanisms of their decision-making processed.			16_techdis-patch_xai_en.pdf
Frugal AI / Green AI	Movement dating from - and revived from - green computing, aiming to reduce the energy use of AI systems. More specifically, a set of techniques that aim to make ML models more efficient, cost-effective, and accessible while maintaining or even improving their performance.	Medium	R&D	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5012869
Responsible AI	Responsible artificial intelligence (AI) is a set of principles that help guide the design, development, deployment and use of AI, building trust in AI solutions that have the potential to empower organisations and their stakeholders. Responsible AI involves the consideration of the broader societal impact of AI systems and the measures required to align these technologies with stakeholder values, legal standards and ethical principles. Responsible AI aims to embed such ethical principles into AI applications and workflows to mitigate risks and negative outcomes associated with the use of AI, while maximising positive outcomes.	Medium	R&D	https://www.ibm.com/think/topics/responsible-ai
AI Ethics	AI ethics is a branch of artificial intelligence focused on ensuring that AI systems are designed, deployed, and governed in ways that are fair, transparent, and aligned with societal values. This field addresses moral, societal, and legal questions arising from the use of AI, such as the impact of biased algorithms, data privacy concerns, and the broader societal implications of automation and decision-making by machines. AI ethics seeks to establish frameworks and principles to guide the development and use of AI technologies responsibly.	Medium	R&D	https://ai-world.eu/concept/ai-ethics

Innovation in AI				
Keyword	**Description**	**Disruption potential[74]**	**Technology maturity[75]**	**Source**
Jailbreak	In the context of Artificial Intelligence, particularly with advanced language models like GPT, Jailbreak refers to methods used by users to bypass the restrictions and safety measures built into AI systems. These models are typically programmed with rules and filters that prevent them from generating harmful, inappropriate, or unauthorised content. A jailbreak occurs when a user manipulates the AI system, through carefully structured inputs or prompts, to make it perform actions or generate outputs that were originally restricted. Jailbreaking can expose vulnerabilities in the model's ability to enforce its safety protocols and ethical boundaries, raising concerns about misuse and the robustness of safety mechanisms.	Low	Deployment	https://ai-world.eu/concept/jail-break
Auditability of an AI system	An AI audit is a systematic examination of an artificial intelligence system to assess its fairness, transparency, security, compliance, and performance. The goal is to ensure that AI operates ethically, legally, and effectively while minimising risks such as bias, discrimination, or security vulnerabilities.	Low	Deployment	https://ec.europa.eu/futurium/en/system/files/ged/auditing-artificial-intelligence.pdf

3.2.2. Taxonomy – Innovation with AI

Innovation with AI - AI capabilities				
Keyword	**Description**	**Economic impact[77]**	**Societal impact[78]**	**Source**
Deep reasoning	Deep reasoning combines deep learning with reasoning for solving complex tasks. It is a critical step toward progress in achieving Artificial General Intelligence.	High	High	https://www.ml-science.com/deep-reasoning Workshop
Reasoning & decision-making	Reasoning problems arise naturally in many applications that interact with the world - commonsense query answering, diagnosis problem solving, planning, reasoning about knowledge in the sciences, NLP, and multi-agent control, to name a few. Reasoning algorithms and knowledge representations form the foundations for theoretical investigations into human-level AI. AI research has led to a range of automated reasoning techniques. These reasoning techniques have given rise to AI algorithms and systems, including SAT, SMT, and constraints solvers as well as probabilistic graphical models, all of which play a key role in critical real-world applications. While large pre-trained systems (such as LLMs) have made impressive advancements in their reasoning capabilities, more research is needed to guarantee correctness and depth of the reasoning performed by them; such guarantees are particularly important for autonomously operating AI agents. In terms of decision making, AI-augmented decision-making is plausible and may initially consist of LLMs applying strategy frameworks and answering questions like an analyst assisting a strategist. Current LLMs can achieve human-comparable performance in	High	High	https://www.cambridge.org/core/books/abs/cambridge-handbook-of-artificial-intelligence/reasoning-and-decision-making/30DFFAF6A0FF92E1D56664302DB41FDE https://aaai.org/wp-content/uploads/2025/03/AAAI-2025-PresPanel-Report-FINAL.pdf https://arxiv.org/pdf/2408.08811

[77] Discussed during the validation workshop
[78] Discussed during the validation workshop.

Innovation with AI - AI capabilities				
Keyword	**Description**	**Economic impact[77]**	**Societal impact[78]**	**Source**
	realistic strategy tasks involving generation and evaluation.			
AI forecasting and prediction	AI is enabled to find complex patterns in data and use them to make predictions about the future course of a time series or the likelihood of certain events that may happen (e.g., defaulting on a credit). AI-driven forecasting offers a wide range of applications in a variety of sectors, allowing businesses to make more informed choices, improve operations, and gain a competitive advantage. Here are some significant uses of AI driven forecasting in various industries: • **Finance**: AI algorithms examine historical price data, market sentiment, and news to anticipate stock price changes, enabling traders and investors in making better investment choices. • **Retail**: Artificial intelligence estimates customer demand for items, assisting merchants in optimising inventory levels and planning production cycles to avoid stockouts and overstock situations • **Energy**: Artificial intelligence-driven projections for solar and wind energy output assist grid operators in managing changes in energy supply and demand, guaranteeing a steady power supply. • **Healthcare**: Using patient data, treatment patterns, and healthcare provider behaviour, AI models estimate demand for pharmaceutical items, improving supply chain operations. • **Manufacturing industry**: AI-driven projections of raw	High	High	Creation of a taxonomy for the European AI Ecosystem a report of the cross-KIC activity "Innovation Impact Artificial Intelligence Chakraborty, M., & Parida, P. (2024, January). Comprehensive Study of AI-Driven Market Forecasting Models and Their Applicability. In 2024 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC) (pp. 1-5). IEEE. Nguyen, T. (2023). Applications of artificial intelligence for demand forecasting. Operations and Supply Chain Management: An

Innovation with AI - AI capabilities				
Keyword	**Description**	**Economic impact[77]**	**Societal impact[78]**	**Source**
	material demand, manufacturing rates, and distribution patterns improve supply chain efficiency and shorten lead times. • **Agriculture**: AI forecasts crop yields by analysing data from weather sensors, soil conditions, and historical yield data, assisting farmers in planning planting and harvesting tactics. Predictive AI models (e.g. regression, time-series models, or complex ML models) learn patterns that precede an outcome, enabling forecasts such as demand, failure, or risk. This capability turns raw data into actionable foresight (e.g. "Which machine is likely to fail next week?").			International Journal, 16(4), 424-434.
AI (scientific) discovery	AI can process huge amounts of data and find, for example, patterns, logical relationships, similarities, groups, or dependencies in them. Artificial intelligence (AI) is being increasingly integrated into scientific discovery to augment and accelerate research, helping scientists to generate hypotheses, design experiments, collect and interpret large datasets, and gain insights that might not have been possible using traditional scientific methods. Generative AI methods can create designs, such as small-molecule drugs and proteins, by analysing diverse data modalities, including images and sequences.	High	High	https://www.nature.com/articles/s41586-023-06221-2 Creation of a taxonomy for the European AI Ecosystem a report of the cross-KIC activity "Innovation Impact Artificial Intelligence
AI planning	AI can look for optimal solutions to problems requiring long sequences of actions in complex environments, where the search space is too vast for human or even exact computer solutions. Examples of AI planning include:	High	High	https://link.springer.com/article/10.1007/s00521-021-06569-4

Innovation with AI - AI capabilities				
Keyword	**Description**	**Economic impact[77]**	**Societal impact[78]**	**Source**
	• Path planning for Unmanned Vehicles. This is particularly relevant for generating obstacle-free flight paths for unmanned aerial vehicles (UAVs) Integration of AI into urban planning (for example, urban planners using AI include sentiment analysis of public comment, computer vision for neighbourhood conditions assessments and school district boundary adjustment, and optimisation processes to improve public transportation routes).			Creation of a taxonomy for the European AI Ecosystem a report of the cross-KIC activity "Innovation Impact Artificial Intelligence
AI creation / generation	AI creation and generation refer to the ability of artificial intelligence to autonomously produce new content, ideas, or solutions across various fields. AI models use machine learning, neural networks, and deep learning techniques to generate text, images, music, videos and code. Examples of AI creation/generation include: • Text generation (AI-powered language models like ChatGPT generate human-like text, answer questions, and assist with writing). • Image generation (AI models like DALL·E or MidJourney create digital artwork, illustrations, and designs). • Video, music and audio generation. • AI for the generation and testing of ideas. Figure 14 provides an overview of model-, system-, and application-level view on generative AI.	High	High	Creation of a taxonomy for the European AI Ecosystem a report of the cross-KIC activity "Innovation Impact Artificial Intelligence McKinsey (2023) The economic potential of generative AI Feuerriegel, S., Hartmann, J., Janiesch, C., & Zschech, P. (2024). Generative ai. Business & Information Systems Engineering, 66(1), 111-126.

Innovation with AI - AI capabilities				
Keyword	**Description**	**Economic impact[77]**	**Societal impact[78]**	**Source**
Natural language processing (NLP)	NLP is a subfield of computer science and artificial intelligence (AI) that uses machine learning to enable computers to understand and communicate with human language. This includes, Automatic Speech Recognition converts spoken language into text. AI models analyse audio waveforms, filter noise, and transcribe words in real time, speech synthesis (Text-to-Speech), sentiment analysis and text summarisation. Below is an overview of the NLP techniques:[79] • Word Tagging ○ Word segmentation ○ Shallow syntax-chunking ○ Named entity recognition ○ Part-of-speech tagging ○ Semantic role labelling ○ Word sense disambiguation • Sentence Parsing ○ Constituency parsing ○ Semantic parsing ○ Dependency parsing ○ Coreference resolution • Text Classification ○ Sentiment analysis ○ Text classification ○ Temporal processing • Text Pair Matching ○ Semantic textual similarity	Medium	High	https://www.ibm.com/think/topics/natural-language-processing https://interoperable-europe.ec.europa.eu/sites/default/files/inline-files/D02.01_Natural%20Language%20Processing%20for%20Public%20Services_4.pdf https://s3.eu-west-1.amazonaws.com/mobidev.biz.cloudfront/2020/03/natural-language-processing-use-cases-for-business-optimisation.pdf

[79] Retrieved from: https://s3.eu-west-1.amazonaws.com/mobidev.biz.cloudfront/2020/03/natural-language-processing-use-cases-for-business-optimisation.pdf

Innovation with AI - AI capabilities				
Keyword	**Description**	**Economic impact[77]**	**Societal impact[78]**	**Source**
	○ Natural language inference ○ Relation prediction • Text Generation ○ Language modelling ○ Machine translation ○ Simplification ○ Summarisation.			
Computer linguistics	Computational linguistics consists of several key components, including syntax, semantics, and pragmatics. Syntax refers to the rules and structure of language, while semantics deals with the meaning of words and sentences. Pragmatics, on the other hand, involves understanding the context and intent behind language use. Computer linguistics enables AI to process, interpret, and potentially "understand" the content of language (e.g., human) as well as interact in such a language with a desired semantic content.	Medium	High	Nilufar, N. (2023). COMPUTER LINGUISTICS FUTURE HISTORY PERSPECTIVE. International Journal of Advance Scientific Research, 3(11), 179-183. Creation of a taxonomy for the European AI Ecosystem a report of the cross-KIC activity “Innovation Impact Artificial Intelligence
Symbolic reasoning	Symbolic reasoning aims at deducing general logic rules from the knowledge graphs	Medium	High	Workshop https://www.sciencedirect.com/science/article/pii/S2666651021000061

Innovation with AI - AI capabilities				
Keyword	**Description**	**Economic impact[77]**	**Societal impact[78]**	**Source**
Conversational AI & virtual assistants	This capability includes chatbots, voice assistants, customer support automation. The following are examples of the operational improvements conversational AI & virtual assistants can have for: • **Customer self-service**. Generative AI–fuelled chatbots can give immediate and personalised responses to complex customer inquiries regardless of the language or location of the customer. By improving the quality and effectiveness of interactions via automated channels, generative AI could automate responses to a higher percentage of customer inquiries. • **Resolution during initial contact**. Conversational AI & virtual assistants can instantly retrieve data a company has on a specific customer, which can help a human customer service representative more successfully answer questions and resolve issues during an initial interaction. • **Reduced response time**. Conversational AI & virtual assistants can cut the time a human sales representative spends responding to a customer by aiding in real time and recommending next steps. • **Increased sales**. Because of its ability to rapidly process data on customers and their browsing histories, the technology can identify product suggestions and deals	Medium	High	https://www.sciencedirect.com/science/article/pii/S0148296323001960 McKinsey (2023) The economic potential of generative AI Mekni, M. (2021). An artificial intelligence based virtual assistant using conversational agents. Journal of Software Engineering and Applications, 14(9), 455-473.

Innovation with AI - AI capabilities				
Keyword	**Description**	**Economic impact[77]**	**Societal impact[78]**	**Source**
	tailored to customer preferences.			
Computer vision	Enables machines to make sense of visual data by recognising objects or persons and understanding the semantics of images or video sequences. This includes Image classification, Object detection, Facial recognition, Optical Character Recognition (OCR), Image segmentation. In terms of specific application areas: Computer vision can analyse images and extract metadata for business intelligence, creating new revenue opportunities and operational efficiencies. • Autonomous vehicle technology uses computer vision to recognise real-time images and build 3D maps from multiple cameras fitted to autonomous transport. It can analyse images and identify other road users, road signs, pedestrians, or obstacles. • In agriculture, computer vision applications automate tasks like monitoring field conditions, identifying crop disease, checking soil moisture, and predicting weather and crop yields. In healthcare, computer vision can support with tumour detection by analysing moles and skin lesions, Automatic X-ray analysis, Symptom discovery from MRI scans.	Medium	Medium	Karn, A. (2021). Artificial intelligence in computer vision. International Journal of Engineering Applied Sciences and Technology, 6(1), 2455-2143. https://www.ibm.com/think/topics/recurrent-neural-networks https://aws.amazon.com/what-is/computer-vision/
Computer audition	Enables machines to process and make sense of signals by recognising, classifying, and understanding audio signals. The application of computer audition includes:	Medium	Medium	Creation of a taxonomy for the European AI Ecosystem a report of the cross-KIC

Innovation with AI - AI capabilities				
Keyword	**Description**	**Economic impact[77]**	**Societal impact[78]**	**Source**
	• Recognition of natural, machine, man-made or musical sounds, genre recognition, query by humming. • Music summarisation and thumbnailing, music annotation. • Audition driven signal processing, synthesis using sound description. • Modelling of musical affect and aesthetics. Computer aided composition and machine improvisation.			activity "Innovation Impact Artificial Intelligence http://dub.ucsd.edu/ComputerAudition/handouts/CAT1h6.pdf

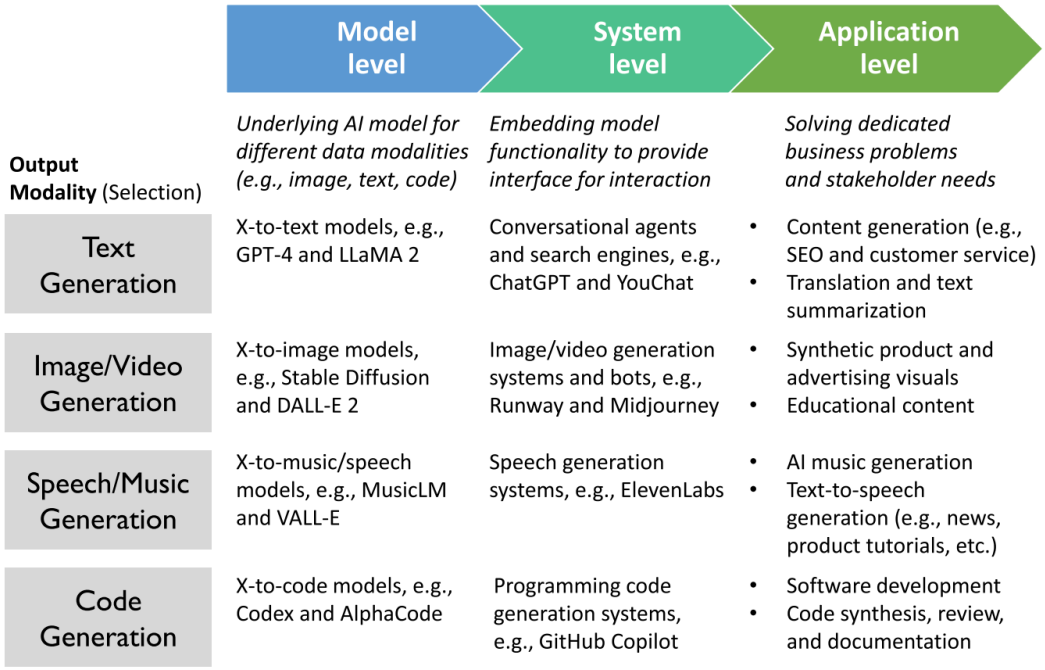


Figure 14 Overview of model-, system-, and application-level view on generative AI. Source: Feuerriegel, S., Hartmann, J., Janiesch, C., & Zschech, P. (2024). Generative ai. Business & Information Systems Engineering, 66(1), 111-126.

3.2.3. Taxonomy – AI application areas

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
AI in science / AI as the engine of innovation	Artificial Intelligence (AI) is revolutionising scientific discovery by accelerating the entire research cycle from knowledge extraction and hypothesis generation to automation of experimentation and verification at an unprecedented speed. This application area includes use cases such as AI research assistants and AI embedded in robotics implementing R&D activities. The bullet points below provide other examples of AI in use cases in science:[82] • Maths proofs and theoretical calculations • Hypothesis generation • Experiment design • Lab robots • Active learning for efficient data sampling • Monitoring, anomaly alerts • Simulation • Super resolution, denoising data • Compression, visualisation, discovering correlations • Uncertainty quantification and propagation • Past decisions on fast hardware • Citizen science • Optimisation of scientific workflows • Re-purpose tools originally developed for AI	High	High	https://aaai.org/ https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/06/artificial-intelligence-in-science_4f3d6efd/a8d820bd-en.pdf https://www.rand.org/pubs/commentary/2024/10/how-ai-can-automate-ai-research-and-development.html

[80] Discussed during the validation workshop.
[81] Discussed during the validation workshop.
[82] Retrieved from: https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/06/artificial-intelligence-in-science_4f3d6efd/a8d820bd-en.pdf

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
	In addition, companies are leveraging AI tools to automate software development, improving efficiency and productivity. AI agents, which can autonomously perform complex tasks, are being developed to streamline R&D processes.			
AI in manufacturing	AI enhances automation by enabling robots to handle repetitive and hazardous tasks. It ensures quality control by using AI-powered vision systems to detect defects in products. Predictive maintenance helps manufacturers avoid unexpected machinery failures. AI also optimises supply chain management by improving inventory tracking and logistics. Key areas where AI has a significant impact include: • **Supply Chain Management**: AI enhances demand forecasting, inventory management, and logistics, reducing forecasting errors. • **Security and safety**: AI/ML can also improve worker and critical equipment safety within factories through intelligent access control systems. It can also be used to mitigate the cybersecurity risks introduced by the ever-increasing number of net-worked devices within a manufacturing plant. • **Warehouse Management**: AI solutions improve warehouse efficiency by predicting demand patterns, optimising inventory, and automating operations. • **Assembly Line Optimisation**: AI algorithms improve precision and efficiency in manufacturing processes, reducing waste and	High	High	https://www.vision4ai.eu/wp-content/uploads/2024/09/Digital_JointSRA_AI-NoE_MHP_20240826_compressed_high.pdf Plathottam, S. J., Rzonca, A., Lakhnori, R., & Iloeje, C. O. (2023). A review of artificial intelligence applications in manufacturing operations. Journal of Advanced Manufacturing and Processing, 5(3), e10159. Patel, D., Lin, S., Shah, D., Jayaraman, S., Ploennigs, J., Bhamidipati, A., & Kalagnanam, J. (2023, September). AI model factory: scaling AI for

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
	ensuring high-quality production. • **AI-based experimentation**: AI and ML can be used for high-fidelity simulations of manufacturing processes and to reduce the need for manual experimentation which saves cost and time • **Predictive Maintenance**: AI helps anticipate equipment breakdowns, minimising downtime and optimising maintenance schedules. Technologies like digital twins, which replicate physical objects in a virtual setting, play a significant role. • **Quality Assurance**: AI improves quality control through computer vision and machine learning, enabling companies to detect defects and maintain consistent product quality. • **Streamlined Paperwork**: AI automates administrative tasks such as invoice processing and quality control checks. • **Demand Prediction**: AI helps predict market demand based on sales data and trends, enabling businesses to optimise inventory and avoid overproduction. • **Connected Factories**: AI-integrated IoT devices create connected factories that enable real-time decision-making, predictive maintenance, and seamless operations. **Process and product design:** Generative design involves the use of AI to explore the design space for a product or			industry 4.0 applications. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 37, No. 13, pp. 16467-16469). Lee, D., & Lin, C. (2024). Universal artificial intelligence workflow for factory energy saving: Ten case studies. Journal of Cleaner Production, 468, 143049.

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
	process based on user-provided requirements.			
AI in public sector	For example, AI-powered chatbots and virtual assistants can handle routine inquiries, improving citizen engagement. Additionally, predictive analytics enables public sector to anticipate demand for public services, optimise budget allocations, and enhance policymaking through data-driven insights. Other examples of use cases include: • **Policymaking**: This is seen particularly in tax administration (e.g., streamlining tax collection processes and preventing tax fraud), budget management (e.g., analysing spending and revenue data to develop and manage budget more effectively, and anticipating budget allocations), public health, welfare benefits, and social services. Although less prominent across the use cases analysed, governments are also using AI to improve policymaking in other sectoral areas, such as transport, environment, security, urban planning, employment, justice, infrastructure, education, foreign affairs, among others • **Service design and delivery**: Governments are also using AI to increase responsiveness in the design and delivery of public services. **Government oversight:** A variety of use cases show that AI can help increase transparency and accountability of government activities, when tested appropriately and used responsively. Tax and procurement agencies, as well as audit institutions have been early	High	High	Tangi, L., Van Noordt, C., Combetto, M., Gattwinkel, D. and Pignatelli, F., AI Watch. European landscape on the use of Artificial Intelligence by the Public Sector, Publications Office of the European Union, Luxembourg, 2022, doi:10.2760/39336, JRC129301. https://oecd-opsi.org/toolkit-navigator/ https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/06/governing-with-artificial-intelligence_f0e316f5/26324bc2-en.pdf

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
	adopters of these technologies, in areas ranging from fraud detection and corruption control to risk management.			
AI in healthcare	AI improves medical imaging by detecting diseases in X-rays, MRIs, and CT scans with high accuracy. It enhances predictive analytics by identifying patterns that indicate potential disease outbreaks or patient deterioration. AI-powered virtual health assistants provide medical guidance and mental health support. Furthermore, AI assists in robotic surgery, enabling precise and minimally invasive procedure. AI also accelerates drug discovery by analysing large datasets to identify potential compounds. It optimises clinical trials by selecting the right patients and improving trial efficiency. AI personalises medicine by tailoring treatments based on genetic data and patient history. Additionally, AI enhances supply chain management by predicting demand and optimising drug.	High	High	https://www.foreseemed.com/artificial-intelligence-in-healthcare https://www.oecd.org/en/publications/ai-in-health_2f709270-en.html Jiang, F., Jiang, Y., Zhi, H., Dong, Y., Li, H., Ma, S., ... & Wang, Y. (2017). Artificial intelligence in healthcare: past, present and future. Stroke and vascular neurology, 2(4).
AI in mobility, automotive and transport	AI enables self-driving vehicles by processing sensor data for navigation and safety. It enhances driver assistance systems with features like adaptive cruise control and lane-keeping assistance. AI predicts vehicle maintenance needs, reducing the chances of breakdowns. Furthermore, AI optimises traffic management by analysing congestion patterns and suggesting better routes. Additional use cases include: • Intelligent Public Transport System.	High	High	https://ai-watch.ec.europa.eu/topics/transport-and-mobility_en Iyer, L. S. (2021). AI enabled applications towards intelligent transportation. Transportation

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
	• Intelligent Safety Management System. • Intelligent Automotive Manufacturing & Logistics System.			Engineering, 5, 100083.
AI and autonomous systems	AI and autonomous systems include, among others, drones, autonomous vessels, delivery robots and humanoids.	High	High	https://www.elspub.com/journals/artificial-intelligence-and-autonomous-systems/home Workshop
Domain-specific GenAI models	Domain-specific LLMs are designed to cater to niche areas with specialised knowledge and vocabulary. The benefits of using domain specific models include: • Security and control. For use cases with sensitive data, agencies can fine-tune a smaller LLM for the specific task and domain. Smaller LLMs can be hosted on the agency's own cloud instance with a cloud service provides. There, the agency can maintain better control of where the data is transmitted and stored. • Cost optimisation. Large-scale LLMs offer great flexibility but can be expensive to run. While smaller, domain-specific models may incur higher costs for training and hosting, they can offer lower compute costs, which may prove more economical for large-volume usage. Examples of such models include BioGPT for biomedical research or FinGPT for financial forecasting.	High	Medium	https://www2.deloitte.com/content/dam/Deloitte/us/Documents/consulting/us-domain-specific-generative-ai.pdf https://medium.com/ai-horizons/generative-ai-episode-9-the-rise-of-domain-specific-large-language-models-0f50ba99d4d9

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
AI in energy intensive industries and energy renewables	AI, for example, improves energy distribution through smart grids, ensuring efficient power management and infrastructure maintenance. It predicts faults in power plants and wind turbines, reducing maintenance costs. AI optimises energy consumption in homes and industries, leading to better efficiency. Furthermore, AI predicts solar and wind energy availability, facilitating better integration of renewable energy sources. AI optimises energy storage systems, ensuring efficient use of stored power, and supports carbon capture, utilisation, and storage (CCUS) by monitoring emissions and improving capture technologies. Additionally, AI aids in, for example, oil and gas exploration by analysing geological data. Figure 16 provides specific use cases of AI in energy intensive industries and energy renewables sector.	High	Medium	https://www.renewableenergymagazine.com/panorama/ai-and-renewable-energy-a-complex-but-20240923 https://www.fdmgroup.com/news-insights/ai-in-energy-sector/ Quest, H., Cauz, M., Heymann, F., Rod, C., Perret, L., Ballif, C., ... & Wyrsch, N. (2022). A 3D indicator for guiding AI applications in the energy sector. Energy and AI, 9, 100167.
AI for Media and Outreach	AI is transforming media and outreach in several ways, enhancing content creation, audience engagement, and distribution strategies.	Medium	High	Workshop
AI in service-based industrial ecosystems (Retail, Culture and Creative Industries,	In retail, AI-driven recommendation systems, chatbots, and inventory optimisation improve customer experiences and streamline operations. In culture and creative industries, AI-powered content generation, virtual reality (VR) experiences, and automated design tools support artists and businesses in creating content. In tourism, AI enhances travel	Medium	High	https://www.europarl.europa.eu/RegData/etudes/BRIE/2020/629220/IPOL_BRI(2020)629220_EN.pdf

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
Tourism, Proximity and social economy)	planning through smart assistants, dynamic pricing, and personalised itineraries, while AI-powered translation tools improve communication for international travellers.			https://www.mckinsey.com/industries/retail/our-insights/llm-to-roi-how-to-scale-gen-ai-in-retail OECD (2024), "Artificial Intelligence and tourism: G7/OECD policy paper", OECD Tourism Papers, No. 2024/02, OECD Publishing, Paris, https://doi.org/10.1787/3f9a4d8d-en. https://www.tsia.com/blog/the-current-and-future-state-of-ai-in-professional-services
AI in education	AI can transform learning and education by offering personalised experiences, offering adaptive learning possibilities that adjust to students' needs. In addition, AI is used in higher education in various ways, from detecting plagiarism and ensuring exam security. Several other use cases include: • Remote digital tutors that use virtual and augmented reality. • AI-driven platforms assess students' skill levels in real time and tailor instructional content to meet individual needs. These systems adapt lessons	Medium	High	https://www.weforum.org/stories/2024/04/future-learning-ai-revolutionizing-education-4-0/ https://onlinedegrees.sandiego.edu/artificial-intelligence-education/

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
	dynamically based on student responses, providing customised pathways to help students master concepts at their own pace. • Assistive technology such as speech recognition software transcribe spoken words into text, helping students with disabilities such as hearing impairments or dyslexia to participate more fully in the classroom by converting speech to text and vice versa. • AI for curricula planning by analysing educational data to identify trends and gaps. This ensures the curriculum remains relevant, comprehensive and aligned with learning objectives by suggesting updates based on the latest educational needs and standards. Virtual 3D Classrooms: The metaverse creates immersive virtual classrooms where students can interact with classmates and teachers.			
AI regulatory technologies	Increased adoption of AI in regulatory technology has significantly expanded its use cases, cutting across banking, securities, insurance, and other financial services, and covering a wide variety of activities. These include identity verification, AML/CFT, fraud detection, risk management, stress testing, micro-prudential reporting, macro-prudential reporting, as well as compliance	Medium	High	https://www.imf.org/en/News/Articles/2021/10/29/sp102921-ai-and-reg-tech
AI in telecommunications	AI optimises network performance to ensure better connectivity. Chatbots and virtual assistants provide round-the-clock customer support. Network operators are using AI models to detect faults, malfunctions, and anomalies within the network, which can cause service	Medium	Medium	https://www.mckinsey.com/industries/technology-media-and-

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
	interruptions. Once an abnormal behaviour is identified, models can be trained to reduce future occurrences. AI also helps detect and prevent telecom fraud. Additionally, AI enhanced 5G deployment by dynamically managing network traffic. A new model of telecommunications is emerging, whereby AI-enabled automation of network management and technology is reducing cost-to-serve. GenAI capabilities are enabling sales and marketing personalisation to drive business growth, with opportunities to diversify revenue streams through AI infrastructure provision and AI-as-a-service (AIaaS) offerings. Natural language capabilities are transforming customer service, unlocking innovations in digital service, including AI-powered retail assistants and natural language chatbots.			telecommunications/our-insights/how-generative-ai-could-revitalise-profitability-for-telcos https://reports.weforum.org/docs/WEF_Artificial_Intelligence_in_Telecommunications_2025.pdf https://www.intel.com/content/www/us/en/learn/ai-in-telecommunications.html
AI in defence	• AI strengthens surveillance and reconnaissance by analysing satellite and drone footage for potential threats (e.g., enhanced geospatial intelligence gathering). It also enhances cybersecurity and resilience from cognitive warfare by detecting and preventing cyberattacks in military networks. AI improves autonomous weapons systems (air, marine and ground vehicles) by increasing precision and decision-making capabilities. It also predicts geopolitical risks and anticipates security threats, aiding in strategic planning. In terms of support of consult, command and control (C3), AI contributes to a decision advantage, promotes and	Medium	Medium	https://www.nato.int/nato_static_fl2014/assets/pdf/2023/3/pdf/stt23-vol2.pdf https://www.strategyand.pwc.com/de/en/industries/aerospace-defense/ai-in-defense.html

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
	enables technologies that can improve decision quality, shorten the decision cycle, and increase information superiority over opponents.			
AI in agriculture	AI supports precision farming by analysing soil and weather conditions to maximise crop yields. It detects plant diseases and pests early, allowing for timely interventions. AI-driven robots automate harvesting, improving efficiency. Additionally, AI monitors livestock health, helping farmers optimise feeding strategies and detect illnesses early. Additional use cases are provided below: • Automated irrigation and feeding systems • Weed and pest management • Crop health monitoring and disease detection • Automated and precision harvesting	Low	Low	https://www.mckinsey.com/industries/agriculture/our-insights/from-bytes-to-bushels-how-gen-ai-can-shape-the-future-of-agriculture https://www.researchgate.net/publication/370213114_Artificial_Intelligence_in_Agriculture
AI in Aerospace	AI enables predictive maintenance, helping to prevent mechanical failures before they occur. It also assists in autonomous navigation for unmanned aerial vehicles (UAVs). AI improves air traffic management by optimising flight routes. Moreover, AI-powered vision systems inspect aircraft parts for defects. Other examples of use cases include: • Aircraft structural damage and durability detection. • Spacecraft guidance and control. • Predictive fault detection and maintenance.	Low	Low	https://dxc.com/us/en/insights/perspectives/paper/the-future-of-ai-in-the-aerospace-industry Abdulrahman, Y., Arnautović, E., Parezanović, V., & Svetinovic, D. (2023). AI and blockchain synergy in aerospace engineering: An impact survey

AI application areas				
Keyword	**Description**	**Economic impact[80]**	**Societal impact[81]**	**Source**
	• Aerodynamic shape optimisation. • Automated welding and aircraft inspection. • Airport security and surveillance. • Auto piloting systems. Air Traffic Management.			on operational efficiency and technological challenges. IEEE Access, 11, 87790-87804. Sutthithatip, S., Perinpanayagam, S., Aslam, S., & Wileman, A. (2021, October). Explainable ai in aerospace for enhanced system performance. In 2021 IEEE/AIAA 40th Digital Avionics Systems Conference (DASC) (pp. 1-7). IEEE.

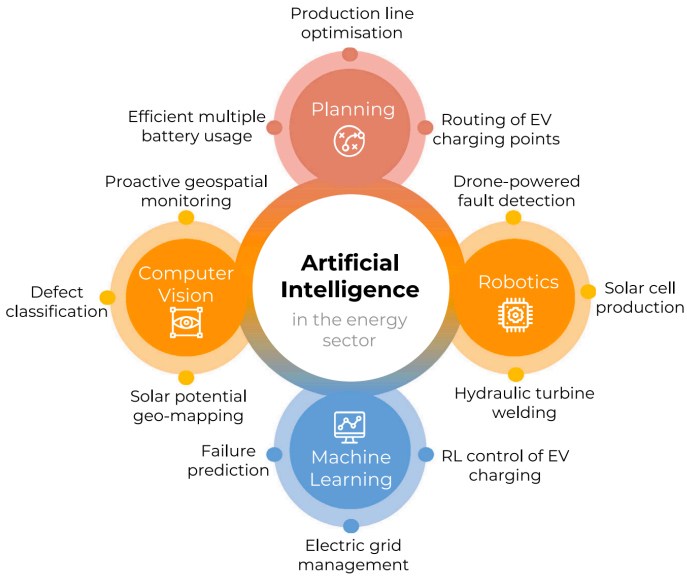


Figure 15 Use cases of AI in energy intensive industries and energy renewables sector. Source: Quest, H., Cauz, M., Heymann, F., Rod, C., Perret, L., Ballif, C., ... & Wyrsch, N. (2022). A 3D indicator for guiding AI applications in the energy sector. Energy and AI, 9, 100167.

4. Annex

4.1. Bibliography

Topic	Selected references
EU Strategy & policy environment	• Data Act[83] • Data Governance Act[84] • Digital Service Act[85] • Digital Markets Act[86] • AI Act[87] • AI Office[88] • AI in Science Strategy[89] • GDPR[90] • Regulation of the Free Flow of non-personal data[91] • EUCS[92] • NIS2 Directive[93] • Communication Artificial Intelligence for Europe[94] • White Paper on AI[95] • Communication on fostering an EU approach to AI[96] • Revised coordinated Action Plan on AI[97] • IPCEI CIS[98] • AI continent action plan[99]

[83] Retrieved from: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32023R2854&qid=1748612245604
[84] Retrieved from: https://eur-lex.europa.eu/eli/reg/2022/868/oj/eng
[85] Retrieved from: https://eur-lex.europa.eu/eli/reg/2022/2065/oj/eng
[86] Retrieved from: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=uriserv%3AOJ.L_.2022.265.01.0001.01.ENG&toc=OJ%3AL%3A2022%3A265%3ATOC
[87] Retrieved from: https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng
[88] Retrieved from: https://digital-strategy.ec.europa.eu/en/policies/ai-office
[89] Retrieved: https://research-and-innovation.ec.europa.eu/research-area/industrial-research-and-innovation/artificial-intelligence-ai-science_en
[90] Retrieved: https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng
[91] Retrieved: https://eur-lex.europa.eu/eli/reg/2018/1807/oj/eng
[92] Retrieved: https://ec.europa.eu/newsroom/cipr/items/713799/en
[93] Retrieved: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32022L2555
[94] Retrieved: https://digital-strategy.ec.europa.eu/en/library/communication-artificial-intelligence-europe
[95] Retrieved: https://commission.europa.eu/system/files/2020-02/commission-white-paper-artificial-intelligence-feb2020_en.pdf
[96] Retrieved: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52021DC0205
[97] Retrieved: https://digital-strategy.ec.europa.eu/en/policies/plan-ai
[98] Retrieved: https://competition-policy.ec.europa.eu/state-aid/ipcei/approved-ipceis/cloud_en
[99] Retrieved from: https://digital-strategy.ec.europa.eu/en/library/ai-continent-action-plan

	• Big Data Value contractual PPP[100] • Invest AI Initiative[101] • AI Innovation Package[102]
Future oriented literature, market reports, policy documents	• AAAI 2025 Presidential Panel on the Future of AI Research. Retrieved from: https://aaai.org/wp-content/uploads/2025/03/AAAI-2025-Pres-Panel-Report-FINAL.pdf • Abioye, S. O., Oyedele, L. O., Akanbi, L., Ajayi, A., Davila Delgado, J. M., Bilal, M., Akinade, O. O., & Ahmed, A. (2021). Artificial intelligence in the construction industry: A review of present status, opportunities and future challenges. • Adra (2024) - Policy paper and technology roadmap GenAI and robotics 4EU • AI Index Report (2024): Annual Report. AI Index Steering Committee, Institute for Human-Centered AI. Authors: Nestor Maslej, Loredana Fattorini, Raymond Perrault et al., Stanford University CA, April 2024. • Artificial Intelligence and the Evolution of AI (Model) Capabilities A Conceptual Primer • Avin, S., Belfield, H., Brundage, M., Krueger, G., Wang, J., Weller, A., Anderljung, M., Krawczuk, I., et al. 2021. Filling gaps in trustworthy development of AI. Science, 374(6573), pp.1327-1329. • Bailey, G., Farinha, J., Mochan, A. and Polvora, A., Eyes on the Future - Signals from recent reports on emerging technologies and break-through innovations to support European Innovation Council strategic intelligence - Volume 1, Publications Office of the European Union, Luxembourg, 2024, doi:10.2760/355, JRC137811. • Beckert, B., & Kroll, H. (2024). Definition of the research and innovation field of" Artificial Intelligence" and approaches to determining quality (No. 88). Fraunhofer ISI Discussion Papers-Innovation Systems and Policy Analysis. • Beraja, M., Kao, A., Yang, D.Y. and Yuchtman, N., 2021. AI-tocracy. National Bureau of Economic Research, November 2021, Working Paper Working Paper 29466. • Boute, R., & Udenio, M. (2021), AI in Logistics and Supply Chain Management. • Brundage, M., Avin, S., Clark, J., Toner, H., Eckersley, P., Garfinkel, B., Dafoe, A., Scharre, et al., 2018. The malicious use of artificial intelligence: Forecasting, prevention, and mitigation. arXiv preprint arXiv:1802.07228. • Bulchand-Gidumal, J. (2020). Impact of Artificial Intelligence in Travel, Tourism, and Hospitality.

[100] Retrieved from: https://bdva.eu/big-data-value-cppp/
[101] Retrieved from: https://ec.europa.eu/commission/presscorner/detail/en/ip_25_467
[102] Retrieved from: https://digital-strategy.ec.europa.eu/en/factpages/ai-innovation-package

- Canals, J. and Heukamp, F. (2020). The future of management in an AI world: Redefining Purpose and Strategy in the Fourth Industrial Revolution.
- CITB (2021). A Challenge and An Opportunity: Artificial Intelligence and in the Construction Industry.
- Creation of a Taxonomy for the European AI Ecosystem: A Report of the Cross-KIC Activity
- CSET (2022): Counting AI Research: Exploring AI Research Output in English- and Chinese-Language Sources. Center for Security and Emerging Technology (CSET). Author: Daniel Chou, Washington D.C.; CSET, July, https://cset.georgetown.edu/publication/counting-ai-research/.
- Draghi, M. (2025). The future of European competitiveness. Retrieved from: https://commission.europa.eu/topics/eu-competitiveness/draghi-report_en
- Dwivedi, Y.K., Hughes, L., Ismagilova, E., Aarts, G., Coombs, C., Crick, T., Duan, Y., Dwivedi, R., Edwards, J., Eirug, A. and Galanos, V., 2021. Artificial Intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy. International Journal of Information Management, 57, p.101994.
- EMI Methodological Report (2024)
- Epoch AI, 'Parameter, Compute and Data Trends in Machine Learning'. Published online at epoch.ai. Retrieved from: 'https://epoch.ai/data/notable-ai-models' [online resource]
- EU-U.S. Terminology and Taxonomy for Artificial Intelligence - Second Edition
- European Commission: Joint Research Centre, Eulaerts, O., Grabowska, M. and Bergamini, M., Weak signals in science and technologies 2024 – Technologies at an early stage of development that could impact our future, Publications Office of the European Union, 2025, https://data.europa.eu/doi/10.2760/6571994
- European Parliament (2023). Artificial Intelligence in the Agri-food Sector: Applications, Risks and Impacts.
- Future of Life Institute, 2023. Pause Giant AI Experiments: An Open Letter. Retrieved from https://futureoflife.org/open-letter/pause-giant-ai-experiments/.
- Hossain, M. N., Rahman, M. M., & Ramasamy, D. (2024). Artificial Intelligence-Driven Vehicle Fault Diagnosis to Revolutionize Automotive Maintenance: A Review.
- Hyder, Z., Siau, K., & Nah, F. (2023). Artificial Intelligence, Machine Learning, and Autonomous Technologies in Mining Industry.
- JRC (2024). Eyes on the Future - Signals from recent reports on emerging technologies and break-through innovations to support European Innovation Council strategic intelligence - Volume 1.2

	• Korinek, A, 2023, DP17923 Language Models and Cognitive Automation for Economic Research, CEPR Press Discussion Paper No. 17923. https://cepr.org/publications/dp17923 • Kumar, I., Rawat, J., Mohd, N., & Husain, S. (2021). Opportunities of Artificial Intelligence and Machine Learning in the Food Industry. • Kurzweil, R., 2022. Superintelligence and singularity. Machine Learning and the City: Applications in Architecture and Urban Design, pp.579-601. • Mission Letter for Commissioner-designate for Startups Research and Innovation (europa.eu) • Munnisunker, S., Nel, L., Diederichs, D. (2022). The Impact of Artificial Intelligence on Agricultural Labour in Europe. • NATO (2024). Science & Technology Trends 2023-2043. Retrieved from: https://www.nato.int/nato_static_fl2014/assets/pdf/2023/3/pdf/stt23-vol2.pdf • Naudé, W., 2021. Artificial intelligence: neither Utopian nor apocalyptic impacts soon. Economics of Innovation and new technology, 30(1), pp.1-23. • OECD (2023): What technologies are at the core of AI? An exploration based on patent data. Authors: Flavio Calvino, Chiara Criscuolo, Hélène Dernis, Lea Samek, OECD Artificial Intelligence Papers, No. 6, OECD Publishing, Paris, https://doi.org/10.1787/32406765-en. • OECD AI policy observatory website • OECD Framework for the Classification of AI systems • Ponce (2020). Collaborative Economy, Tourist Accommodation and Their Impact in the Context of Sustainable Urban Development: Is Artificial Intelligence a Possible Answer? • Savills News. (2024). European data centre power capacity projected to rise to approximately 13,100 MW by 2027. Available at: https://www.savills.be/research-and-news/savills-news/362723/european-data-centre-power-capacity-projected-to-rise-to-approximately-13-100-mw-by-2027 • Shane Legg, Marcus Hutter (2007). Universal Intelligence: A Definition of Machine Intelligence. Minds & Machines, 17:4, pages 391-444. • Stauffer et al., 2023. Existential Risk and Rapid Technological Change: Advancing Risk-informed Development, United Nations Office for Disaster Risk Reduction. • Yampolskiy, R.V., 2019. Predicting future AI failures from historic examples. Foresight, 21(1), pp.138-152. • Yampolskiy, R.V., 2020. Unpredictability of AI: On the impossibility of accurately predicting all actions of a smarter agent. Journal of Artificial Intelligence and Consciousness, 7(01), pp.109-118.
Sectoral studies & selected report on	• European Commission (2024) Communication on boosting startups and innovation in trustworthy artificial intelligence

trend development for AI, data & robotics	• ECA (2024) Special report 08/2024: EU Artificial intelligence ambition – Stronger governance and increased, more focused investment essential going forward • JRC (2024) Three Challenges to Secure AI Systems in the Context of AI Regulation • EC (2023) Foresight on Demand: Artificial General Intelligence, Issues and Opportunities • ECA (2023) Audit preview 02/2023: Artificial intelligence in the EU • JRC (2023) Analysis of the preliminary AI standardisation work plan in support of the AI Act • European Commission (2022), Artificial Intelligence: A European Perspective on Ethics, Data, and Technology • OECD (2023) AI in Business and Government: Market Adoption and Trends • World Economic Forum (2023) AI for the Future of Work and Society • Stanford University (2023) AI Index Report: Measuring the State of Artificial Intelligence • JRC (2022) AI Watch: AI for enhancing Robotics • JRC (2022) AI Watch: AI uptake in Manufacturing • JRC (2022) AI Watch: Estimating AI Investments in the European Union • JRC (2021) AI Watch: AI Uptake in Smart Mobility • JRC (2021) AI Watch. Beyond pilots: sustainable implementation of AI in public services • JRC (2021) National strategies on Artificial Intelligence: A European perspective • HE funded Forging Hub project • Brightlio (2024) Data Center Stats https://brightlio.com/data-center-stats/ • Cloud Hacks (2024) Federated Learning: A Paradigm Shift in Data Privacy and Model Training • European Commission (2023) Foresight on Demand: Artificial General Intelligence, Issues and Opportunities • European Commission (2025) *Second annual report on key findings from the European Monitor of Industrial Ecosystems (EMI)* Accompanying the document Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions "The 2025 Annual Single Market and Competitiveness Report". Retrieved from: https://single-market-economy.ec.europa.eu/document/download/f8d5d980-eba4-4fd7-90ad-0d01c3e2855f_en?filename=SWD_2025_12_F1_OTHER_STAFF_WORKING_PAPER_EN_V5_P1_3878248.PDF

	• European Investment Bank (2021) Artificial Intelligence, Blockchain and the Future of Europe: How Disruptive Technologies Create Opportunities for a Green and Digital Economy • Expert Investor (2022) Data centre boom expected in Europe • PwC (2017) Sizing the price: What's the real value of AI for your business and how can you capitalise? • World Economic Forum (2021) Data-Driven Economies: Foundations for Our Common Future • World Economic Forum (2021) Data-Driven Economies: Foundations for Our Common Future • World Economic Forum (2019) AI's Contribution to the Global Economy Will Bypass that of China and India by 2030, to Reach $15.7 Trillion • Skypoint (n.d.) The Death of the API for Analytics: Why Data Sharing is the Future • European Monitoring of Industrial Ecosystems, uptake of AI in 14 industrial ecosystems
Websites	• https://adr-association.eu/news/adra-releases-policy-paper-and-technology-roadmap-genai-and-robotics-4eu • https://aiworld.eu/ • https://artificialintelligenceact.eu/article/3/ • https://commission.europa.eu/topics/eu-competitiveness/draghi-report_en • https://digital-strategy.ec.europa.eu/en/library/commission-publishes-guidelines-ai-system-definition-facilitate-first-ai-acts-rules-application • https://digital-strategy.ec.europa.eu/en/library/eu-us-terminology-and-taxonomy-artificial-intelligence-second-edition • https://digital-strategy.ec.europa.eu/en/policies/ai-factories • https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Use_of_artificial_intelligence_in_enterprises • https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Business_economy_by_sector_-_NACE_Rev._2 • https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/wp-call/2023-2024/wp-7-digital-industry-and-space_horizon-2023-2024_en.pdf • https://engineering.princeton.edu/news/2025/01/06/ai-slashes-cost-and-time-chip-design-not-all • https://epoch.ai/data/notable-ai-models#faq • https://media.licdn.com/dms/document/media/v2/D4D1FA-QEKkwTxKK78WQ/feedshare-document-pdf-analyzed/B4DZTapqukGkAY-/0/1738835159295?e=1742428800&v=beta&t=oFEg_BxU5XwPthNcb-STap5Z0P8ln1JeJDoDWUZL6paE

	• https://monitor-industrial-ecosystems.ec.europa.eu/sites/default/files/2023-12/EMI%20Methodological%20Report.pdf • https://oecd.ai/en • https://www.ai-dapt.eu/project/ • https://www.eca.europa.eu/ECAPublications/SR-2024-08/SR-2024-08_EN.pdf • https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/tech-forward/digital-twins-and-generative-ai-a-powerful-pairing • https://www.mediahuis.ie/app/uploads/2024/04/HAI_AI-Index-Report-2024.pdf • https://www.morganstanley.com/articles/ai-cybersecurity-new-era • https://www.nvidia.com/en-us/high-performance-computing/hpc-and-ai/ • https://www.weforum.org/stories/2023/12/ai-fobo-jobs-anxiety/

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4.3. Taxonomy building methodology and next steps of the project

The taxonomy has been built based on the following steps:

1. **Review of established taxonomies on AI technology:**

- The team reviewed the list of AI taxonomies / typologies and frameworks focusing on their scope, granularity and application areas.

2. **Constitution of the literature corpus for keyword extraction:**

- As direct follow-up from the identified literature sources, our team selected a list of literature sources constituting the core background (training dataset) for AI technologies, ensuring a good representativity of AI technology development and adoption across sectors.
- The data was collected in a single database to easily identify duplicate values and ensure that the sources identified capture AI technology.

3. **Establishing a taxonomy:**

- Our data science expert then deployed text mining techniques on the core list of sources (training data). This allowed the team to provide a robust definition and taxonomy (list of keywords) for AI technology development and adoption. More specifically:
    - We used a large language model (LLM) to extract AI-related keywords from each report. When reports included predefined keyword lists, those were extracted in full.
    - We consolidated all keywords from the reports and removed duplicates or very similar terms to ensure consistency.
    - We calculated the frequency of each keyword across reports to provide an initial indication of importance.
    - The LLM was also used to cluster similar keywords, helping to form a preliminary structure.
    - Experts reviewed the aggregated keyword list, filtered and rephrased terms as needed, and refined the taxonomy to ensure clarity and relevance.
- Thematic experts then reviewed the list of keywords (taxonomy) and assessed if the definition and keywords provided enable to distinguish AI technology from others in a relevant manner. The keywords' list was then refined according to experts' feedback.

4. **Updated list of keywords and their categorisation based on the training data**

- The data science expert updated the list of keywords from additional sources.
- The output from this stage consisted in additional keywords in the database specifying new identified keyword and their category.

5. **Human validation of the classification**

- The results of the final list of keywords and their categories were reviewed by the team.
- This led to a manual reclassification of keywords.

In terms of the next steps, the methodology for assessing AI adoption across European industrial sectors (Task 2) will be structured around the AI taxonomy developed in Task

1. This taxonomy will serve as the foundation for data collection and analysis by providing a comprehensive list of AI-related keywords and concepts relevant to each sector. Using this structured approach, the Consortium will create queries and search parameters that are aligned with sector-specific applications and terminology, ensuring consistency and comprehensiveness in data gathering across all sources.

The taxonomy will initially be applied uniformly to search across data sources such as patent databases, job advertisements, and industry reports. However, as data collection progresses, Consortium will adapt it to incorporate sector-specific AI applications and terminology relevant to each industry. By the end of Task 2, the taxonomy will be refined to a higher level of precision, creating a framework that is both shaped by and adapted to the diverse ways AI is implemented across sectors. This dynamic approach ensures that the taxonomy remains comprehensive and relevant, capturing the full scope of AI technologies as they pertain to each NACE sector.

To get a comprehensive view of AI adoption, the Consortium will employ a multi-source data collection strategy that combines both quantitative and qualitative inputs. These data sources include:

- Sources of quantitative data: EU Survey on ICT Usage and E-commerce in Enterprises (carried out by Eurostat), formal patent tracking, bibliometric analysis, job advertisements, and Crunchbase data on sector-specific funding and market activity. These sources will offer sector-specific data on AI presence, innovation, investment trends, and workforce demand.
- Sources of qualitative data: industry reports, new literature, Horizon Scanning sources, expert interviews, and a targeted expert survey will provide context on sectoral AI applications, challenges, and future opportunities. This qualitative data will be essential to fill gaps, especially where quantitative data might be limited or lack contextual depth.

Using the evolving taxonomy as a guiding structure, the Consortium will develop queries and data collection strategies that are adaptable, enabling to add new terms and sectoral nuances as they emerge. Given the variety of data sources and sector sizes, classification and normalisation are crucial for consistent and comparable analysis. Each data point will be categorised according to the AI taxonomy and its associated NACE sector, creating a dual classification by technology and sector. This will allow to organise data effectively and ensure that insights reflect both the specific AI technology and the industry context in which it is applied.

Normalisation will address the differences in sector size and economic impact to allow for accurate cross-sector comparisons. For example, factors such as the total number of companies, market size, workforce size, or startup numbers will be adjusted for using Eurostat's *Business Economy by Sector - NACE Rev. 2* publication as a baseline.[103]

With the data classified, normalised, and aligned to an improved taxonomy, a scoring system will be implemented to assess AI adoption levels within each sector. Sectors will be evaluated against defined criteria that reflect key factors such as prevalence and centrality of AI in each sector. Scores will allow stakeholders to understand sectoral performance in AI adoption relative to each other, supporting cross-sectoral benchmarking. Finally, the Consortium will visualise each sector's score and relative position in AI adoption.

4.4. Validation workshop and consulted stakeholders

The following organisations have been consulted:

[103]Retrieved from: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Business_economy_by_sector_-_NACE_Rev._2

- ADRA
- CEPS
- EIT Digital
- European Commission
- IJS
- INESC TEC
- IPC
- JRC
- KU Leuven
- OECD
- SINTEF
- Tallin University of Technology
- TNO

Screenshots from the validation workshop whiteboards are provided below.

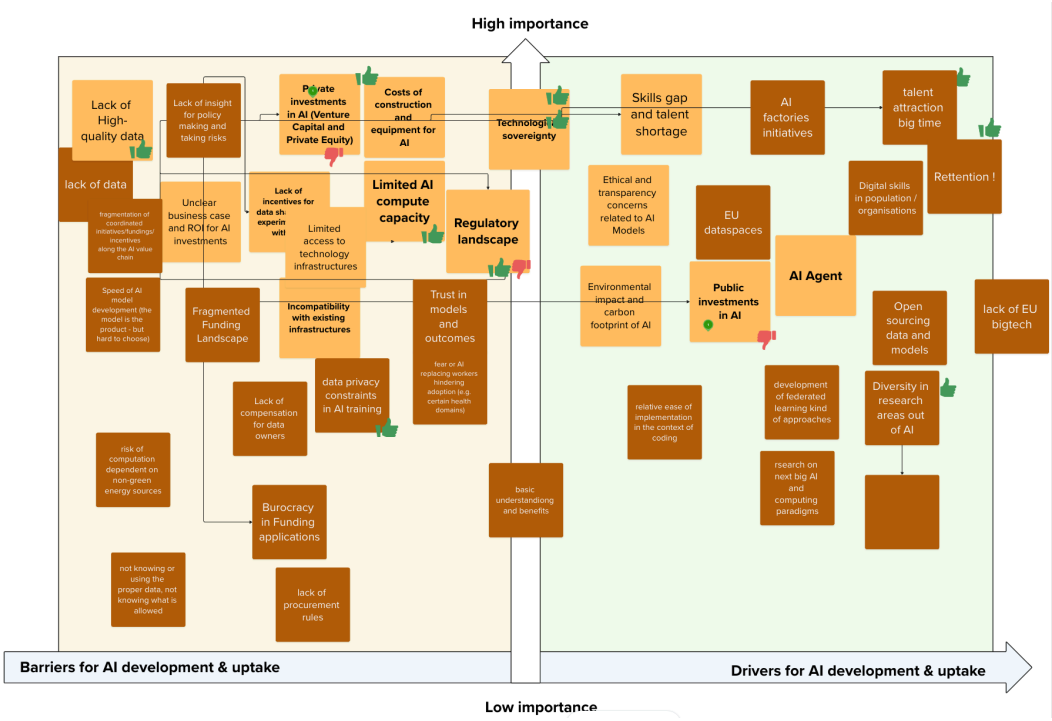


Figure 16 - - Barriers and Drivers for AI Development and Adoption. Source: Technopolis Group, Validation Workshop – 18th of March 2025

The level of importance for each driver and barrier was classified based on expert judgment. This assessment was conducted during the validation workshop, where a group of domain experts shared their views and insights. The classification emerged through consensus among the experts present. The workshop served as a platform for collective discussion, allowing experts to evaluate each factor's significance based on their professional experience and contextual knowledge.

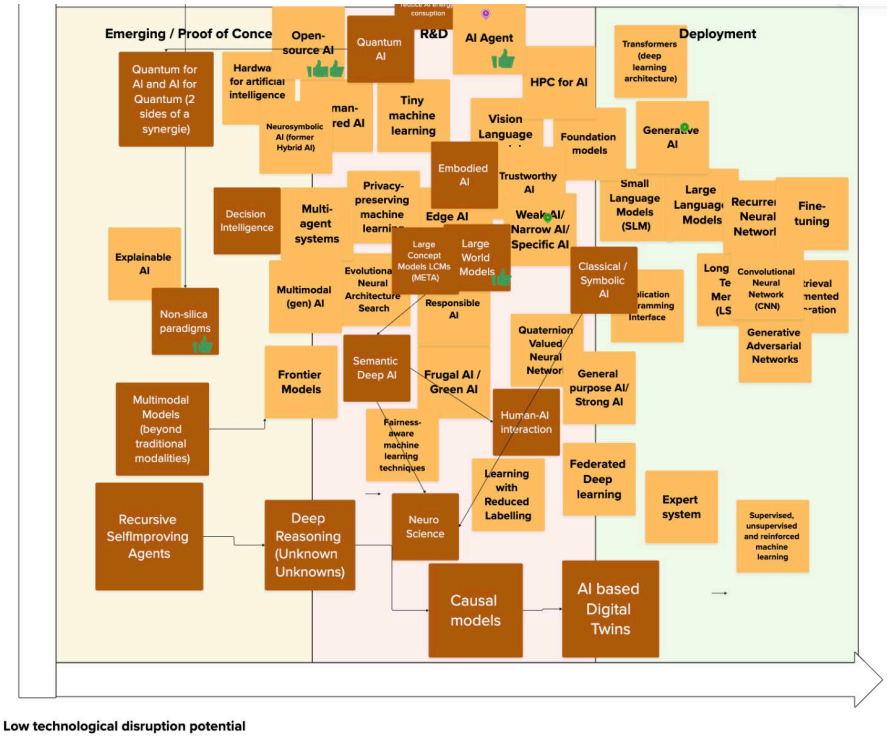


Figure 17 - Taxonomy building: emerging and critical technology trends. Source: Technopolis Group, Validation Workshop – 18th of March 2025

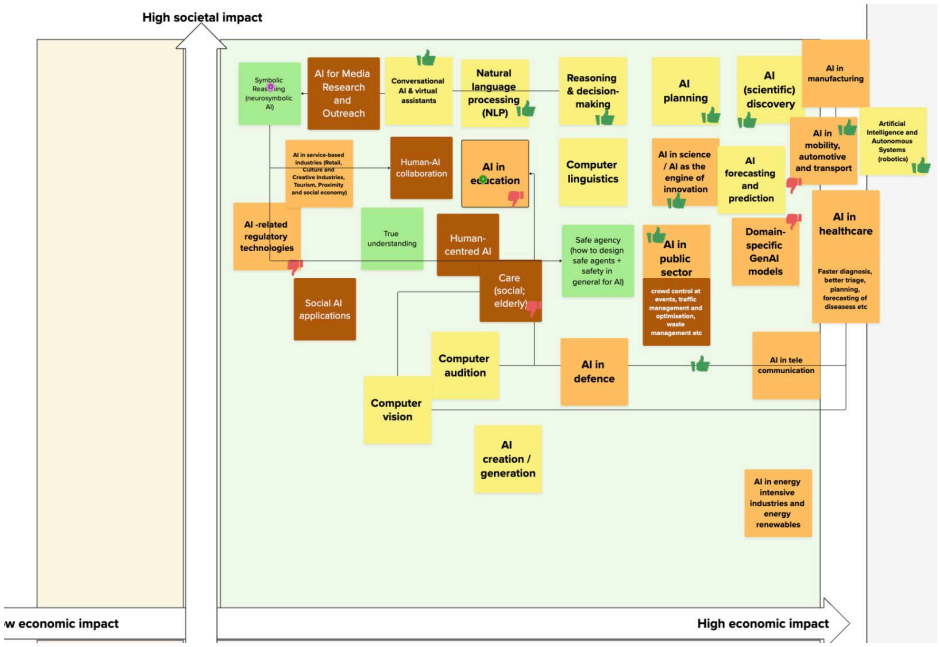


Figure 18 - AI applications and capabilities. Source: Technopolis Group, Validation Workshop – 18th of March 2025

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This report is the first deliverable of the study: “The Futures of Artificial Intelligence: Implications for Europe’s R&I Ecosystem”. It outlines the context and methodology for the creation of a taxonomy of Artificial Intelligence (AI). In this report, the trends of AI generation and adoption are listed and analysed, as well as the barriers for development and uptake of AI, emerging technologies and cross-technological synergies. The taxonomy is then described in detail, integrating innovations in AI, AI capabilities and application areas. This taxonomy is designed to reflect the guidelines on AI systems definition, and to facilitate future research in AI systems.

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